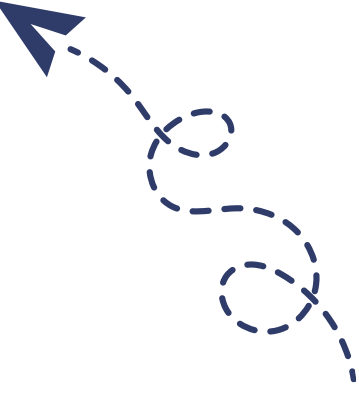




قناة تانية تمريضيانو على التليجرام



Arab Republic of Egypt

Ministry of Health and Population

Central Administration for Human Resource Development

General Administration of Technical Health Education

Nursing Administration

Secondary Nursing School

Internal Medicine, Fevers, Dermatology and Venereology

Second Year

2016-2020

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Internal Medicine, Fevers, Dermatology and Venereology

Second Year
Curriculum Framework

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Introduction

Learning without the skill of thinking is useless, while thinking without knowledge is fraught with danger.

Indeed, knowledge gained through the experience of dealing with and caring for patients is one of the most important means of learning. However, students may struggle when trying to extract information from large books. Therefore, we aimed to present this work in the clearest, most comprehensive, and concise scientific material, keeping pace with the latest medical information, hoping that this book will be a helpful friend for students of the Secondary Technical Nursing School.

Finally, we can only thank everyone who contributed to the production of this work.

We hope to God that this book will be your guide and means to serving patients.

And God is the Grantor of success

The Authors

Immune Response

Infectious microbes and humans are bound in a complex and ongoing struggle. The evasive and distinctive mechanisms of microbes that cause disease are countered by the human immune system. When the microbes prevail, disease occurs; on the other hand, when the immune system succeeds, the human survives.

❖ Methods of the body's resistance to disease:

Natural Immunity (Innate Immunity)	Acquired Immunity (Adaptive Immunity)
1. Skin and Mucous membrane.	1. Mucosal immune system (Peyer's patches)
2. Complement	2. Antibodies
3. Neutrophils, Macrophages	3. Cellular components (B, T Lymphocytes)

❖ **Acquired immunity is divided into:**

➤ **First Type: Active Acquired Immunity**

This occurs when the human immune system is exposed to an (antigen), leading to the production of (Antibodies) and also the formation of memory cells like (T-Lymphocytes Based memory).

Examples: Immunization with live attenuated or dead microbes or their toxins (Toxoid).

➤ **Second Type: Passive Acquired Immunity**

This occurs when a person receives ready-made antibodies or immune cells from another person.

Example: A person receiving Immunoglobulin.

First Chapter
Heart and Blood Vessels Diseases
Angina Pectoris

➤ Definition:

It is severe pain beneath the breastbone (sternum), and the pain of angina is described as squeezing, pressing, or burning in the chest.

➤ Causes:

The occurrence of angina is linked to narrowing of the coronary arteries (there are 2 main coronary arteries), which is known as atherosclerosis, and it occurs frequently in:

1. Excessive obesity.
2. Diabetes.
3. Increased triglycerides and cholesterol in the blood.
4. High blood pressure.
5. Hereditary causes.
6. Excessive consumption of animal fats like butter - oil... frequently.
7. Excessive smoking.
8. Lack of physical activity.

➤ What is atherosclerosis:

It is the deposition of fats and cholesterol beneath the inner wall of the arteries, leading to narrowing of the artery and a deficiency in oxygen.

Symptoms:

- 1 - Affects men more than women.
- 2 - Presence of pain under the breastbone; the pain increases with exercise - eating - upon exposure to severe cold - and also occurs at rest. The pain spreads especially to one of the arms - the wrist - the shoulder - the teeth.
- 3- The pain is accompanied by some symptoms like sweating - and nausea.

➤ Diagnosis of Angina Pectoris:

1. Regular ECG or stress ECG.
2. Nuclear scan.
3. Use of cardiac catheterization.

➤ Treatment:

1. Reassuring the patient.
2. Smoking cessation - Avoid nervous tension.
3. Weight reduction - exercising - controlling blood pressure - controlling blood sugar.
4. Reducing animal fats - and salt in food.
5. Use of arterial vasodilators like Nitroglycerin (Isordil 5 mg) sublingually.
6. Administer medications to lower cholesterol (Ator) - Aspirin 150 mg, 2 tablets after lunch.
7. Arterial cardiac catheterization - to dilate the arteries or place a stent or surgical procedure to replace the arteries (CABG).

Myocardial Infarction

Definition:

It is severe chest pain resembling angina pectoris pain but longer in duration and greater in intensity, accompanied by profuse sweating - nausea - and severe fear for the patient.

Causes:

Same causes as angina pectoris - but there is a complete blockage in the coronary artery leading to death of a part of the heart muscle.

Symptoms:

1. Severe pain under the breastbone.
2. The pain spreads to the left arm - the shoulder and the neck.
3. Sometimes the pain is a burning sensation in the stomach.
4. Sweating - nausea - vomiting - and the extremities are cold.

Complications:

- 1- Heart failure - pulmonary edema.
- 2- Arrhythmias (heart rhythm disturbances).
- 3- Spread of the clot to other parts.

Diagnosis:

- 1- ESR - CBC (Complete Blood Count).
- 2- Cardiac enzymes . CPK
- 3- Regular ECG.
- 4- Chest X-ray – Heart Ultrasound (Echocardiogram).

Treatment:

- 1- Admission to the cardiac care unit (CCU).
 - 2- Humidified oxygen - Morphine ampoule.
 - 3- IV Heparin ampoule is given / 4 hours.
 - 4- Aspirin tablets or anticoagulant like Dindevan are given.
 - 5- In some cases, the drug Streptokinase is used - to dissolve the clot.
 - 6- Performing cardiac catheterization and carrying out dilation or angioplasty of the coronary arteries
 - 7- Ensuring control of sugar level - lipids - and blood pressure.
 - 8- Nitroglycerin ampoule (Tridil).
 - 9- Treating complications like arrhythmias and heart failure.
-

Hypertension

Definition:

Hypertension is diagnosed when the blood pressure reading is 140/90 or more in three different readings.

Causes:

- 1- Primary (Essential) Type, which represents more than 90% of cases, and its causes (Obesity - Hereditary factors - High salt intake in food - Smoking).
- 2- Secondary Type, which is due to:
 - a- Kidney diseases like infections - renal artery stenosis.
 - b- Thyroid disorders (hyperthyroidism).
 - c- Adrenal gland disorders.
 - d- Brain tumors.
 - e- Use of medications like Cortisone.

Symptoms:

- 1- There are no specific symptoms in most cases. (High blood pressure is often discovered by chance).
- 2- General fatigue.
- 3- Headache - blurred vision.
- 4- Epistaxis (Nosebleed) - Haematuria (Blood in urine)

Complications:

- 1- Cardiac hypertrophy and heart failure.
- 2- Arrhythmia (heart rhythm disturbance).
- 3- Brain hemorrhage.
- 4- Acute or chronic kidney failure.
- 5- Visual impairment due to retinal affection

Diagnosis:

- 1- Complete Blood picture - blood sugar measurement - Serum urea test - Creatinine - Cholesterol - Triglycerides.
- 2- Fundus examination.
- 3- Echocardiogram (Ultrasound on the heart).
- 4- Chest X-ray.
- 5- ECG.

Treatment:

- 1- Treating the causes leading to high blood pressure.
- 2- Avoiding stress and psychological anxiety, and the use of sedatives is permitted.
- 3- Reducing excess weight.
- 4- Reducing salt in food.
- 5- Practicing simple exercise like walking.
- 6- Smoking Cessation.
- 7- Treatment with medications along with periodic blood pressure monitoring. Examples:
 - Epilat
 - Inderal (Propranolol).
 - Capoten (Captopril).

Heart Failure

Definition

It is the failure of the heart to pump an adequate amount of blood to the various tissues of the body.

Note:

- The heart pumps about 5 liters/minute.
 - The heart beats about 70 - 100 beats/minute.
-

Causes:

- 1- High blood pressure.
 - 2- Blockage in the main coronary arteries.
 - 3- Fibrosis in the heart muscle (Cardiomyopathy).
 - 4- Stenosis or regurgitation in the main valves like the mitral or aortic.
-

Symptoms:

- 1- Shortness of breath and difficulty breathing during exercise.
 - 2- Attacks of shortness of breath during sleep, accompanied by profuse sweating, cyanosis of the face, and sputum with foam and blood.
 - 3- Frequent cough.
 - 4- Swelling of the feet (Edema).
 - 5- Liver enlargement and abdominal pain.
-

Diagnosis:

- 1- Chest X-ray.
 - 2- ECG.
 - 3- Echocardiogram (Heart Ultrasound).
-

Treatment:

- 1- Treating causes like high blood pressure - arterial stenosis - valve dilation.
- 2- Complete rest in bed in a semi-sitting position.
- 3- Administer Humidified oxygen.
- 4- Give diuretics, e.g., Lasix - Aldactone.

- 5- Digoxin (Lanoxin) tablets.
 - 6- Aminophylline injection IV slowly.
 - 7- Reducing salt in food.
-

Cardio Pulmonary Resuscitation C.P.R

Causes of Sudden Cardiac Arrest:

- Sudden cardiac arrest leads to a drop in circulation and cessation of breathing, causing loss of consciousness. If it continues for more than 5 minutes, it causes permanent damage to brain cells.
- First aid must be performed quickly and easily to avoid complications.

Causes of Cardiac Arrest:

- 1 - Coronary artery insufficiency - high blood pressure - mitral or aortic valve disorder.
- 2 - Electric shocks - drowning - suffocation.
- 3 - Doses of drugs like Morphine - Cocaine - Heroin.
- 4 - During surgical operations and when giving anesthesia.
- 5 - Accidents.

What are the signs of cardiac and respiratory arrest:

- 1 - Loss of consciousness.
- 2 - Absence of pulse - drop in blood pressure.
- 3 - Breathing cessation
- 4 - Change in facial color.
- 5 - Dilated pupil.

Main steps for cardiopulmonary resuscitation:

- 1 - The victim is placed on their back, preferably with a hard surface behind their back.
- 2 - An IV cannula is inserted.
- 3 - An (Laryngeal airway) Air Way device is placed in the mouth to ensure air reaches the lungs.

4 - Artificial respiration is performed by mouth or using an Ambu bag.

5 - Cardiac massage is performed at a rate of 4 cardiac compressions to 1 breath.

6 - If possible, connect the patient to a monitor.

7 - The medications required to be available during resuscitation are:

Adrenaline ampoule - Xylocaine (Lidocaine) ampoule.

Atropine ampoule.

Calcium ampoule.

Glucose solution: 5%.

Sodium bicarbonate solution.

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Second Chapter

Chest Diseases

Bronchial Asthma

Definition:

Attacks of difficulty in breathing resulting from recurrent narrowing of the bronchi and the presence of viscous secretions in them.

Causes:

Bronchial asthma can occur at any age:

1. Allergy: Attacks occur upon exposure to allergens or fumes like house dust, bird feathers, animals, fungi in the air, pollen in spring.
Also, some foods like fish, milk, eggs, chocolate, or some food additives like flavor enhancers, preservatives.
2. Allergy to some medications like Aspirin and its derivatives - Painkillers.
3. Allergy in children due to exercise.
4. There is a type of bronchial asthma that occurs in adults and is considered of unknown cause.

Symptoms:

1. Chest tightness - difficulty breathing.
2. Wheezing sound during inhalation and exhalation.
3. Severe sweating - and exhaustion in severe cases.
4. Cough with phlegm.
5. Cyanosis of the face and palms - drop in blood pressure - increased pulse.
6. The attack lasts from hours to even days.

Diagnosis:

1. chest X-ray.
2. Pulmonary function tests.
3. Blood gases ABG.
4. Complete blood picture.

Treatment:

- 1- Bronchodilators are given via inhaler or nebulizer device (like Ventolin - Atrovent...).
- 2- An IV line is established with saline solution + bronchodilators like Aminophylline are given.
- 3- Corticosteroid preparations are given IV in severe cases.
- 4- Adrenaline injections are given (in children and with great caution) subcutaneously.
- 5- Broad-spectrum antibiotics are given.
- 6- Humidified oxygen at a rate of 3-4 liters/minute.
- 7- Intensive care and use of a mechanical ventilator in cases of respiratory failure.

Pulmonary Tuberculosis

Definition:

The microbe of pulmonary tuberculosis (TB bacillus) has two types and is called Mycobacterium Tuberculosis.

- a) From humans (Human).
- b) From cattle (Bovine).

Factors of TB infection (Epidemiology):

- 1- Age: common in the period from 5-15 years. (But it affects all ages).
- 2- Affects males and females.
- 3- Crowded places – narrow – and humid with poor ventilation (like prisons – schools – contact with animals on farms – slaughterhouses).
- 4- Poor nutrition – diabetic patients – AIDS patients – and cancer.
- 5- Contacts of pulmonary TB patients.

Symptoms of Pulmonary TB:

1. General weakness – weight loss – loss of appetite – fever – and profuse sweating at night.
2. Dry cough then cough with phlegm and sometimes bloody sputum.
3. Difficulty breathing.
4. Chest pain.
5. Spread of the disease to other parts of the body like the intestines – heart – kidneys – brain – skin – lymph nodes.

Diagnosis:

1. chest X-ray.
2. Sputum analysis for TB microbe (3 times).
3. Blood picture – ESR.
4. Tuberculin test under the skin and read after 72 hours.
5. Culture and sensitivity for the TB microbe.
6. Bactec Test or PCR test.
7. Monitoring treatment (Rifampicin) using liver functions ALT, AST.

Treatment of TB:

1. Complete rest in bed.
 2. Maintaining a healthy diet and taking vitamins like vitamin B6.
 3. Smoking Cessation.
 4. Isolating the patient when their sputum is positive for bacteria (considering them a source of infection).
 5. Controlling sugar level or treating accompanying diseases.
 6. Medical treatment in the form of tablets: INH (Isoniazid) – Ethambutol – Rifampicin – Injection Streptomycin – Pyrazinamide.
 7. Medical treatment in the form of tablets: INH (Isoniazid), Ethambutol, Rifampicin; injections: Streptomycin; and Pyrazinamide
 8. The patient must be monitored daily (taking vital signs – patient's weight – appetite ...).
-

Third Topic

Pneumonia

Definition

It is an acute inflammation of one or more lobes of the lung, caused by inhalation of the pneumococcal microbe (Pneumococci).

Methods of infection:

- 1- Through air contaminated with bacteria.
 - 2- Through contaminated tools (in hospitals), like oxygen masks - laryngeal tubes.
-

Symptoms:

- 1- Sudden fever - and chills.
 - 2- Increased pulse - and increased respiratory rate (normal 12-16 times/minute).
 - 3- Dry cough then becomes accompanied by mucoid sputum mixed with blood sometimes.
 - 4- Feeling of difficulty breathing.
 - 5- Acute pain in the chest wall.
-

Complications:

- 1- Spread of the disease between lobes or across lungs.
 - 2- Occurrence of respiratory failure.
 - 3- Spread of the disease to the heart - kidneys - brain - joints.
 - 4- Acute heart failure.
-

Diagnosis:

- 1- Chest X-ray.
 - 2- Blood picture - ESR.
 - 3- Sputum analysis for ordinary microbes.
-

Treatment:

- 1- Bed rest.
 - 2- Administer humidified oxygen.
 - 3- Give antipyretics like Aspirin – Paramol (Paracetamol).
 - 4- Broad-spectrum antibiotics given by IV injection.
 - 5- Give IV fluids when necessary.
 - 6- Treat heart failure.
 - 7- Monitor vital signs.
-

Chronic Obstructive Pulmonary Disease (COPD) **And Pulmonary Emphysema**

Causes:

- 1- Smoking in all its forms like cigarettes - cigars - hookah (shisha) - and others.
 - 2- Exposure to smoke - and polluted dust.
 - 3- Microbial infection, which accompanies the disease.
-

Symptoms:

- 1- Persistent cough for at least several months each year for two consecutive years.
 - 2- The cough is accompanied by viscous sputum in large quantities and may contain blood or pus.
 - 3- Feeling of difficulty breathing and wheezing in the chest.
 - 4- Cyanosis of the body, especially the lips, with increased respiratory rate.
-

Diagnosis:

- 1- Chest X-ray.
 - 2- Pulmonary function tests.
 - 3- Performing blood gases.
-

Treatment:

- 1- Complete cessation of smoking.
- 2- Avoiding dust and smoke that pollute and irritate the airways.
- 3- Cough suppressant medications.
- 4- Broad-spectrum antibiotics.

5- Humidified oxygen (the quantity determined by the doctor)+ Bronchodilators are given via inhalation like sprays (Ventolin - Clenil).

6- Aminophylline tablets to dilate the airways.

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Part Three

Gastrointestinal Bleeding

Introduction:-

Despite modern advancements in the diagnosis and treatment of gastrointestinal bleeding cases, they still represent the most common cases encountered by emergency room doctors in hospitals. The mortality rate resulting from gastrointestinal bleeding reaches nearly 10%.

Causes:-

(1) Causes of Upper Gastrointestinal Bleeding.

- Esophagitis, Esophageal varices.
- Esophageal cancer.
- Gastritis or Gastric ulcer.
- Duodenitis or Duodenal ulcer.
- Gastric cancer.

(B) Causes of Lower Gastrointestinal Bleeding.

- Colitis or Ulcerative colitis.
- Angiodysplasia.
- Hemorrhoids.
- Colon polyps.
- Diverticulitis.
- Ischemic Colitis.
- Colon Cancer.

Symptoms:-

A patient with acute gastrointestinal bleeding complains of feeling (dizziness - weakness - increased heart rate - sweating - difficulty breathing).

➤ Bleeding appears in one of three forms:-

(A) Hematemesis (Hematemesis)

The patient vomits blood or coffee-ground-like material and the source of hematemesis is the upper gastrointestinal tract.

(B) Black, tarry stool (Melena)

Its source is the upper gastrointestinal tract, the small intestine, and up to the mid-transverse colon. It is blood that has been blackened by the action of gastric acid in the stomach or the effect of bacteria in the case of the colon.

(C) Rectal bleeding (Hematochezia):

Its source is usually the lower gastrointestinal tract, and sometimes, when bleeding is severe from the upper gastrointestinal tract.

Diagnosis :-

1- Clinical examination of the patient :-

(Measuring blood pressure, pulse, and temperature)

2- Blood draw and performing :-

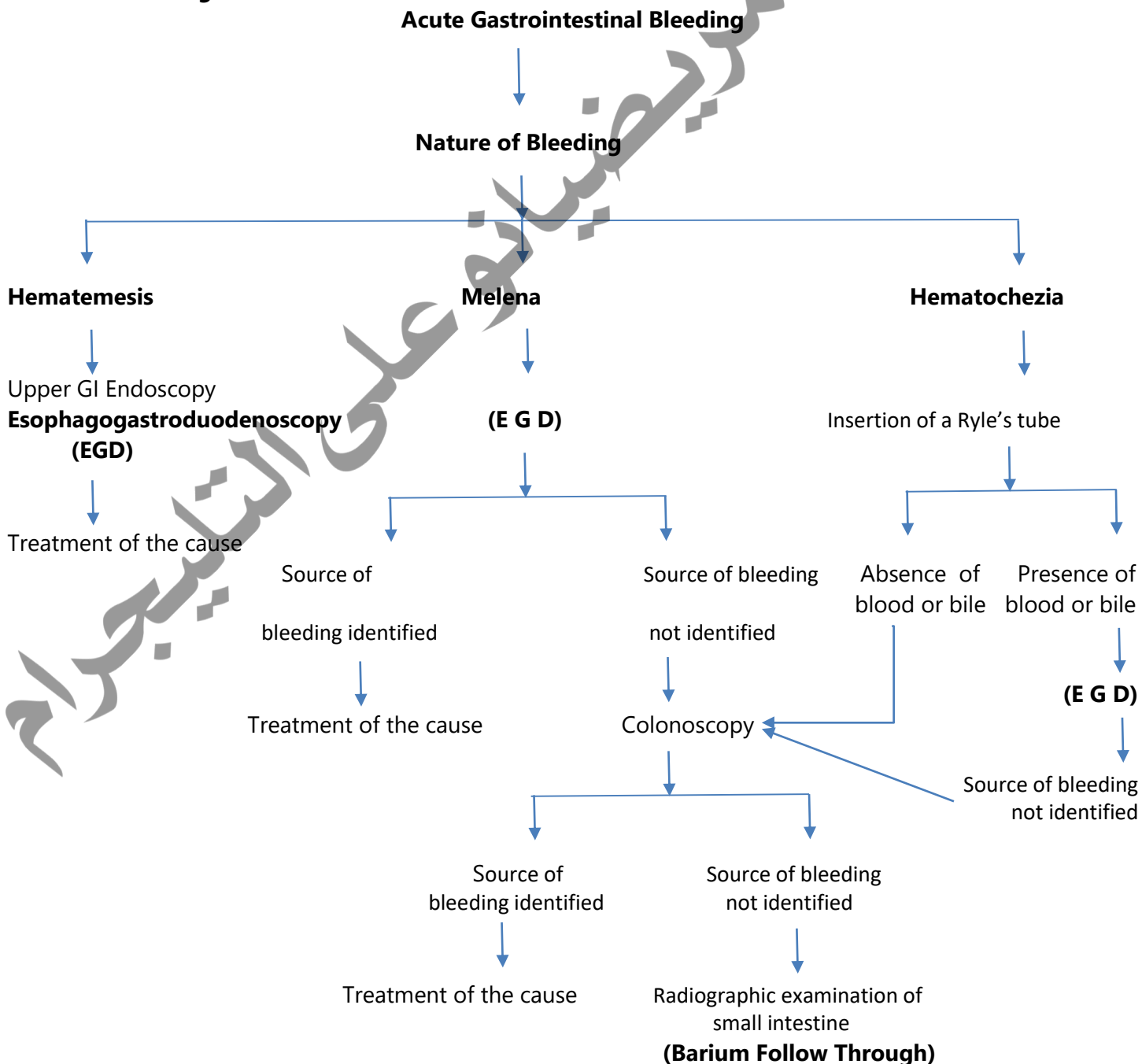
- Complete blood count (CBC).
- Blood type and crossmatch.
- Kidney function (Blood Urea Nitrogen (BUN) and Creatinine).
- Liver function tests, liver enzymes (SGOT, SGPT), Serum albumin.
- Prothrombin Time (P.T.)
- Serum electrolytes.

Treatment :-

1- Restoring or replacing lost blood as follows :-

- Inserting 2 large-bore IV catheters.
- Giving the patient Isotonic Saline solution.
- Giving the patient Packed Red Blood Cells.
- Giving a cirrhotic patient (Fresh Frozen Plasma - Platelets).

2- Treating the cause.



Peptic Ulcer

Peptic and duodenal ulcer is one of the clinically stubborn diseases characterized by a weakness in the mucous membrane lining the wall of the stomach and duodenum. In the past, doctors attributed the cause to increased secretion of gastric acid in the stomach as well as increased secretion of Pepsin, hence the famous saying "No acid, no ulcer" (No Ulcer without acid).

The causes leading to ulcer formation:-

- **Helicobacter Pylori bacteria.**
- Use of painkillers and anti-rheumatic drugs (Anti-rheumatics).
- Zollinger-Ellison syndrome.
- Viral infections.
- Radiation or chemotherapy
- Alcohol and spicy food.
- Nervous stress and nature of work.

➤ **The role of Helicobacter bacteria in the formation of stomach and duodenal ulcers.**

Infection occurs from contamination of food and drink with fecal waste.

Proliferation of Helicobacter pylori in the gastric wall.

↓

Chronic bacterial infection of the stomach wall

↓

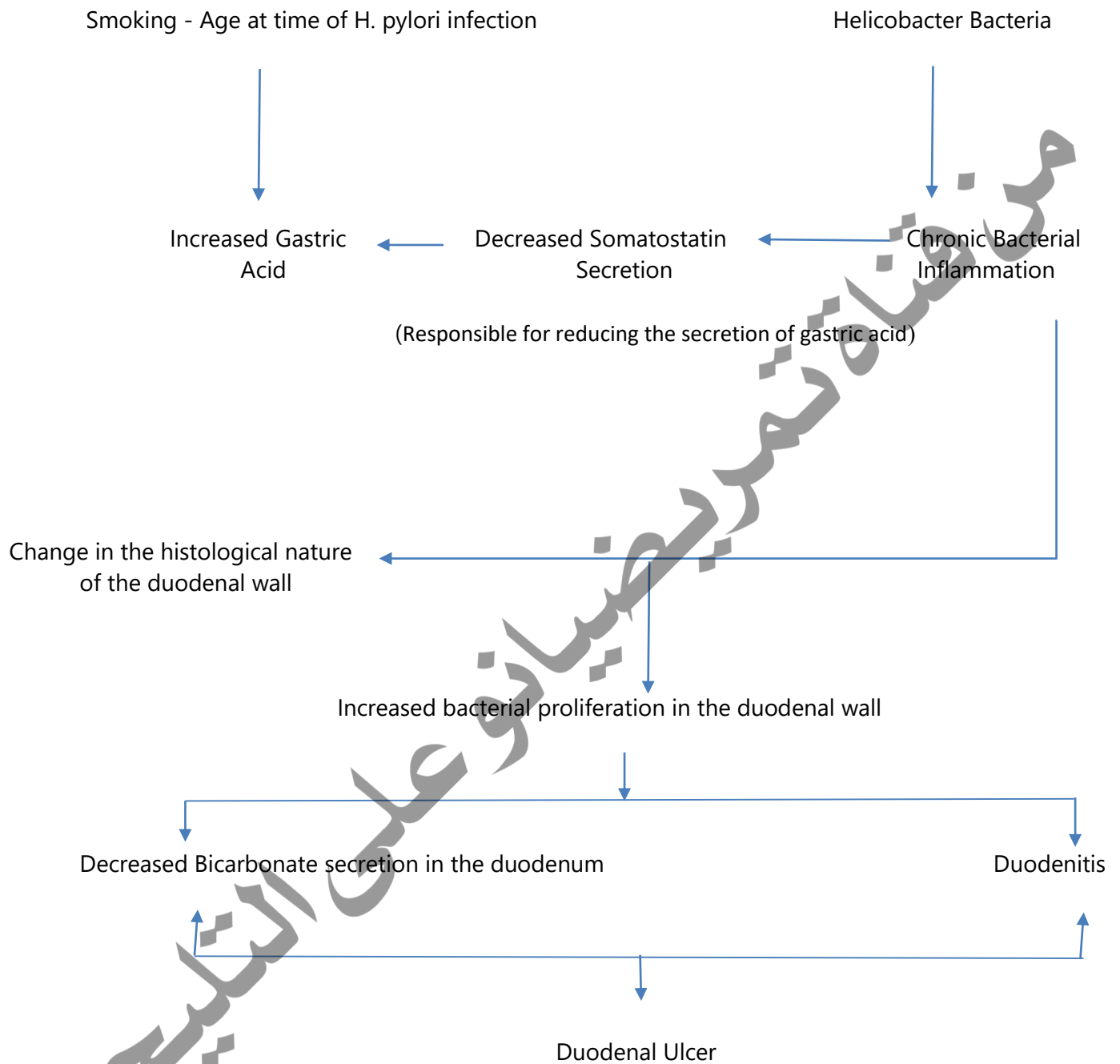
Bacterial toxins (Cag-A - Vac A)

↓

Increased severity of inflammation.

↓

Gastric ulcer



Symptoms :-

- 1- Abdominal pain that is relieved by food in case of duodenal ulcer and increases in case of gastric ulcer.
- 2- Unnoticed anemia due to bleeding from the ulcer and iron deficiency.
- 3- Nausea or vomiting.
- 4- Loss of appetite and weight loss.

Complications -:

- 1- Hematemesis.
- 2- Obstruction of the pyloric opening between the stomach and duodenum.
- 3- Ulcer perforation

Diagnosis -:**Diagnosis of Ulcer:**

- 1- Fiberoptic endoscopy of the upper gastrointestinal tract.

For diagnosing Helicobacter pylori infection:

- 1- Measuring the level of antibodies against the bacteria in the blood.
- 2- Taking a sample from the ulcer via endoscopy and examining it under the microscope.
- 3- Urea Breath Test.
- 4- Stool Antigen Test.

Treatment:

- 1- Easily digestible food free of spices.
- 2- Avoidance of smoking and alcohol.
- 3- Drugs that inhibit acid secretion before meals (Proton Pump Inhibitors). Example: Lansoprazole 30 mg.

Treatment for Helicobacter pylori for 14 days:

- 1- Lansoprazole 30 mg twice daily.
 - 2- Amoxicillin 1 gm twice daily.
 - 3- Clarithromycin 500 mg twice daily.
-

Acute Diarrhea

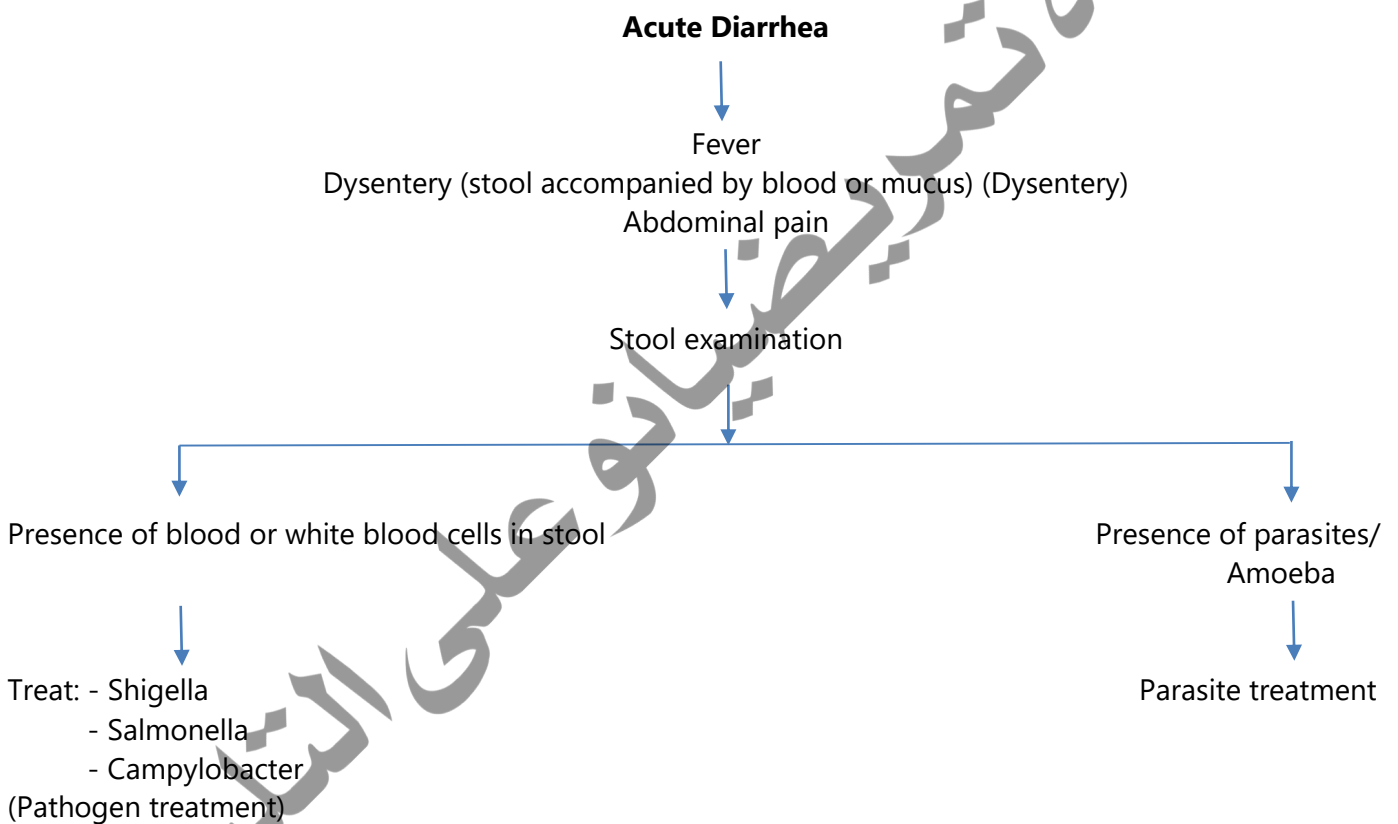
Causes

1- Invasive and destructive microbes affecting the colon wall.

- Shigella
- Salmonella
- Campylobacter Jejuni
- Entamoeba Histolytica

2- Non-invasive microbes (secrete toxins)

- Cholera (Vibrio Cholera)
- E. Coli



Treatment:

1- Replacing lost fluids.

2- For Shigella: Ciprofloxacin 500 mg tablet twice daily for 5 days.

3- For Amoeba: Metronidazole 750 mg every 8 hours for 5 days.

Fourth Topic Viral Hepatitis

Chronic Viral Hepatitis

Acute Viral Hepatitis

Acute Viral Hepatitis

Causes:-

Infection occurs due to infection with one of the hepatitis viruses. The most famous are:-

- **HAV**
- **HBV**
- **HDV**
- **HCV**
- **HEV**

H : Hepatitis

V : Virus

Symptoms:-

1- The symptoms of acute viral hepatitis are similar to flu symptoms such as feeling of severe fatigue, slight fever, muscle and joint aches.

2- These symptoms are accompanied by nausea or repeated vomiting.

3- After that, yellowing of the eyes (sclera) and sometimes the skin also appears (Jaundice) and the patient or those around him may notice this yellowing, and a change in urine color (like tea) may also be noticed.

4- The symptoms of acute hepatitis last about 4-6 weeks.

5- These symptoms do not differ by the type of virus, but the severity of their appearance differs according to the patient's age and also according to the type of virus.

(A) For example, with HAV infection, symptoms are mild in children and may be limited only to digestive system discomfort and nausea, but in adults the symptoms are more severe and clear.

(B) With HCV infection, in most cases there are no symptoms and the patient does not complain of anything.

(C) Sometimes with HAV infection, after the patient feels better, jaundice reappears again (Relapsing Hepatitis) and this relapse occurs within two months of the start of the viral infection.

Modes of Transmission :-

Method	A	E	B	D	C
Contaminated Food	√	√	-	-	-
Contaminated Drink	√	√	-	-	-
Among family members	-	-	√	√	√
Needle sticks	-	-	√	√	√
Drug addicts	-	-	√	√	√
Blood transfusion	?	?	√	√	√
Dialysis	-	-	√	√	√
Via sex	-	-	√	√	√
From pregnant mother to child	-	-	√	√	√

If blood carrying the HEV or HAV virus is transfused from an asymptomatic patient during the virus's incubation period, then the infection is transmitted through blood.

Viral Hepatitis (A)

Acute Hepatitis (A)

HAV infection is transmitted from the patient to the healthy person through eating and drinking water contaminated with the patient's feces and urine.

- In non-urban areas (Low hygienic status) and due to environmental pollution, the rate of HAV infection is higher than in urban areas.
- In non-urban areas, HAV infects children in their first years of life and the infection in most cases has mild symptoms and the child is diagnosed with **Acute Viral gastroenteritis**.
- In urban areas, HAV infects older children and adults and the symptoms are clear and complete.
- In all cases, the patient acquires lifelong immunity "Life Long" against HAV and does not get infected again.

Complications :-

In a very small percentage, acute hepatitis (A) may lead to acute liver failure and coma.

- Acute hepatitis (A) never leads to chronic liver inflammation.
-

Viral Hepatitis (B)

Acute Hepatitis (B)

Infection with this virus is transmitted from carriers to healthy individuals via blood transfusion and blood products

- This virus infects children and adults.
 - Children are infected from mothers carrying the virus (vertical transmission), while infection in adults is through other ways such as contaminated surgical instruments - tattoos - acupuncture needles - injected drugs - sexual relations - contaminated endoscopes - contaminated dialysis machines.
-

Complications

- The danger of this virus lies in its ability to transition the patient from the stage of acute hepatitis to chronic hepatitis, which leads to liver cirrhosis and liver tumors.
- The rate of this transformation is high in children (about 90%) and in adults about 10%.

Acute Viral Hepatitis (D)

Acute Hepatitis (D)

Infection with HDV only occurs in patients infected with HBV. The infection may be simultaneous with HBV, and the patient in this case has acute hepatitis caused by both viruses, or the infection may be sequential, first: HBV then HDV(Superinfection), and the patient in this case has chronic hepatitis caused by Chronic hepatitis B and then gets infected with HDV causing an acute flare Acute hepatitis D (Acute D on top of Chronic B).

- HDV is similar to HBV in modes of transmission, disease progression, and complications.

Acute Viral Hepatitis (C)

Acute Hepatitis (C)

- HCV is transmitted from patients to those infected through contaminated blood and its products, similar to HBV in this regard, but it is rarely transmitted through sexual intercourse or from mother to her newborn child.
- Egypt is considered one of the countries with the highest rates of HCV infection in the world.
- The danger of this virus lies in the fact that 50% of those infected progress to chronic hepatitis, which leads to liver cirrhosis and tumors.

Acute Viral Hepatitis (E)

Acute Hepatitis (E)

Infection with HEV resembles HAV in terms of modes of transmission and its inability to cause chronic liver diseases, but if a pregnant woman is infected with it, it carries a risk of acute liver failure.

Diagnostic Methods

Laboratory Investigations

Clinical Examination

(1) Liver Function Tests **Liver Function Tests (L . F . T.)**

(A) Liver Enzymes :-

Liver enzymes rise significantly in acute hepatitis, possibly reaching ten times or more.

- ALT
- AST
- ALk .P (*Alkaline Phosphatase*)
- GGT (*Gamma-Glutamyl Transferase*)

(B) Bilirubin level :-

The bilirubin level in the patient's blood rises to more than 1mg/dl and more.

(C) Prothrombin Time :-

S. Albumin

These are not affected except in severe cases and with the occurrence of acute liver failure.

(2) Investigations for detecting viruses

A) For Acute Hepatitis (A)

- anti _ HAV _ antibodies
- There are two types:
- Anti _ HAV _ IgM ..They are present with acute infection and disappear with patient recovery.
- Anti _ HAV _ IgG .. They appear after the disappearance of the first type and last for life, indicating previous infection with this virus.

B) For Acute Hepatitis (B)

- Surface antigen of the virus - HBs Ag.
- Its presence indicates that the patient is suffering from an acute infection or a chronic infection (chronic Hepatitis B) or (Acute Hepatitis B).
- Antibodies to the surface antigen of the virus (Anti HBs antibodies).
These antibodies appear at the time of the disappearance of HBs Ag in the patient with acute Hepatitis B. This means the development of immunity and patient recovery.
- HBe Ag - Indicates that the patient is a strong source of infection
- anti HBe Antibodies - Indicates that the patient is not a source of infection.
- anti HBe IgM - Present in case of acute hepatitis.
- anti HBe IgG - Appear with the disappearance of IgM and indicates the end of the acute hepatitis phase, or it may only mean that the patient has had a previous HBV infection, or it may mean the presence of chronic hepatitis if accompanied by the continued presence of HBs Ag.
- There is also the P.C.R. test that detects the presence of the virus itself.

C) For Acute Hepatitis (C)

- We infer the presence of HCV in the patient with acute hepatitis through the P.C.R. test that detects the presence of the virus itself.
- As for the antibodies Anti HCV Antibodies, They appear late after 3-5 weeks from the time of acute infection.

D) For Acute Hepatitis (D)

- Anti HDV IgM - indicate the presence of acute liver infection.
- Anti HDV IgG - indicate the presence of chronic liver infection.

E) For Acute Hepatitis (E)

- Anti HEV IgM - indicate the presence of acute liver infection.
- Anti HEV IgG - indicate that the patient has had a previous infection with HEV.

Methods of Prevention from Viral Hepatitis

- 1- Vaccination of children and adults against hepatitis (A, B).
- 2- Attention to personal hygiene.
- 3- Attention to public hygiene.
- 4- Not using other people's personal items such as scissors, razors, hairbrushes, and toothbrushes.
- 5- Ensuring the sterilization of surgical instruments and endoscopes.

❖ **Treatment:**

A	Not needed
B	IFN - LAM .. ADV
D	IFN - LAM .. ADV
C	IFN - .. RIB
E	Not needed

- IFN = Interferon
 - LAM = Lamivudine
 - RIB = Ribavirin
 - ADV = Adefovir
-

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Chronic Viral Hepatitis

-If acute hepatitis persists for more than 6 months, the patient enters the stage of chronic hepatitis.

-This happens for those infected with the hepatitis virus (B) and (C).

Symptoms :-

- 1 - Feeling of constant fatigue.
- 2 - Noticing yellowing of the eyes (Jaundice) which may be accompanied by a change in urine color at sometimes.
- 3 - The patient may feel an increase in abdominal size due to enlargement of the liver and spleen.
- 4 - Often the patient does not complain of anything at all, and the disease is only discovered when performing viral tests for travel or when undergoing surgery.

Complications :-

- 1 - Occurrence of liver cirrhosis with increased portal vein pressure (Portal Hypertension) and the appearance of varices (Esophageal Varices) and abdominal ascites.
- 2 - With the deterioration of liver function, the patient becomes susceptible to hepatic coma.
- 3 - Occurrence of liver tumors.

Diagnostic Methods :-

- 1 - Liver functions:
 - Elevated liver enzymes.
 - Elevated bilirubin level.
 - Low albumin.
 - Prolonged prothrombin time.

- 2 - Viral investigations :-

A) In case of chronic hepatitis (B) (CHB):

- HBs Ag, anti HBc IgG, PCR.

B) In case of chronic hepatitis (C) (CHC):

- anti HCV, PCR.

- 3- Abdominal ultrasound:

It detects the presence of enlarged spleen and liver, detects the presence of liver cirrhosis and elevated portal pressure, and is used to diagnose the presence of ascites and liver tumors.

Treatment:

- Interferon alpha drug or Lamivudine or Adefovir to treat CHB.
 - Interferon alpha drug and Ribavirin to treat CHC.
-

Schistosomiasis or Bilharziosis

- Schistosomiasis is widespread all over the world, especially in Africa and South America and many countries in Asia.
- Schistosomiasis is named after the German doctor Theodor Bilharz who discovered the worm in 1851 AD.
- Human infection occurs after contact with one of the stages of the schistosoma parasite (the Cercaria stage) when swimming in fresh water.
- This stage of the schistosoma parasite (Cercaria), when it comes into contact with human skin, penetrates it and then moves to the blood vessels of the urinary bladder (in the case of *S. haematobium*) or the mesenteric vessels (in the case of *S. mansoni*).
- After a period, the Cercaria transforms into a complete worm that lays eggs which travel with the blood and leave it, penetrating tissues causing inflammation and then fibrosis in the affected tissues.

Symptoms of Schistosomiasis:

- 1- In the case of (*S. haematobium*) (urinary tract infection):
 - Bloody urine – urinary tract infections – bladder and ureter stones – hydronephrosis and renal failure – and may progress to malignant tumors in the bladder and ureters.
- 2- In the case of (*S. mansoni*) (intestinal and hepatic infection):
 - Diarrhea and blood in stool – anemia – liver cirrhosis – spleen enlargement – elevated portal vein pressure – esophageal varices – bleeding esophageal varices.

Diagnosis:

- 1- Urine analysis (repeated) .. eggs.
- 2- Stool analysis (repeated) .. eggs.
- 3- Rectal snip with microscopic examination . . eggs.
- 4- Antibodies against schistosoma in the blood (Anti bilharzial antibodies).
- 5- Presence of schistosoma eggs in the bloodstream

Treatment: Praziquantel.

Ascites

- **Definition:**
It is the accumulation of serous fluid in the peritoneal cavity of the abdomen, resulting in an increase in abdominal size.
- **Causes:**
 - 1- Acute and chronic heart failure.
 - 2- Renal failure.
 - 3- Liver failure
 - 4- (T.B.) Tuberculosis
 - 5- Peritoneal metastases secondary to malignant neoplasms.

Cirrhotic ascites

❖ **Ascites is a manifestation of deteriorating liver function in patients with liver cirrhosis.**

❖ **Its symptoms:**

- 1- Steady increase in abdominal size.
- 2- May be accompanied by swelling of the feet.
- 3- The patient complains of fatigue and difficulty breathing due to abdominal pressure on the diaphragm.
- 4- The patient complains of loss of appetite, early satiety with a feeling of fullness, and indigestion.
- 5- Protrusion of the umbilicus outward and in severe cases it may rupture with fluid leakage, necessitating surgical intervention.
- 6- Congestion and protrusion of the abdominal wall veins, appearing as tortuous tubes on abdominal examination.

❖ **Treatment:**

- 1- Restrict salt completely from all types of food.
- 2- Use diuretics.
- 3- Monitor the patient's weight, as the response to diuretics is known from the steady decrease in the patient's weight in addition to the disappearance of foot swelling and reduction in abdominal size.

❖ **Complications:**

❖ **The presence of ascites in the abdomen may lead to serious pathological conditions which are:**

- (1) Spontaneous Bacterial Peritonitis.
 - (2) Refractory ascites (presence of ascites unresponsive to treatment).
 - (3) Hepatorenal syndrome (disturbance in kidney function).
-

Hepatic Coma (encephalopathy)

Hepatic coma is one of the most dangerous complications of deteriorating liver function and results from a dysfunction in the higher brain centers due to deteriorating liver function.

- Hepatic coma occurs in liver patients suffering from:
 - Acute hepatic failure
 - Chronic hepatic insufficiency

1) Hepatic coma in case of acute hepatic failure :-

- It is the result of rapid (acute) and severe deterioration in liver function.
- **Symptoms :-**
 - Preceding the patient's entry into a coma may be a state of agitation where the patient is very nervous and prone to violence.
 - This is followed by the patient entering a state resembling natural sleep during which the patient responds to external stimuli.
 - With further deterioration in liver function, the patient enters a complete coma and does not respond to any external stimuli.
- **Treatment :-**
 - The patient needs intensive medical care.
 - In some cases, the patient may need a liver transplant from a living donor.
 - Some patients may need a procedure similar to kidney dialysis where a liver dialysis machine is used to perform liver functions until the patient's liver improves. This treatment line is expensive and not widely available worldwide.

2) Hepatic coma in case of chronic hepatic failure :-

- It is the result of the continuous deterioration of liver functions, and there may be external factors that contribute to its occurrence, which are:
 - 1- Infectious diseases of the respiratory or urinary system or the peritoneal membrane of the abdomen.
 - 2- Bleeding from esophageal or gastric varices or bleeding from a gastric ulcer.
 - 3- Overuse of diuretics.
 - 4- Excessive drainage of peritoneal fluid.
 - 5- Excessive protein intake.
 - 6- Severe constipation or diarrhea.
 - 7- Use of sedative and hypnotic drugs.

Symptoms:-

(A) First stage (or pre-coma stage where there is no accompanying disturbance in the patient's level of consciousness): in which there is:

- Changes in the patient's personality (easily irritated - nervousness - behaving like children - extreme happiness or extreme sadness - frequent forgetfulness).
- Sleep disturbances in the patient: (insomnia - sleeping during the day and lack of sleep at night).
- Upon examining the patient: (hand tremors - a distinctive mouth odor (Fetor Hepaticus) - change in the patient's voice with slow speech).

(B) Coma stage (may not be reached if the patient responds to treatment in the previous stage).

A gradual deterioration in the patient's consciousness occurs, starting with patient's loss of sense of place and time and inability to recognize people or even identify themselves, followed by mild coma resembling normal sleep and then deep coma.

- **Treatment:** Begins at any stage of coma and includes:
 - 1- Prohibiting animal protein from the patient's diet.
 - 2- Discontinuation of diuretics.
 - 3- Using Lactulose syrup.
 - 4- Administration of Neomycin.
 - 5- Performing repeated enemas while observing stool color.
 - 6- Using antibiotics.
 - 7- Performing endoscopy for the esophagus and stomach in case of blood in the stool or suspicion of bleeding varices, where they are injected or ligated to stop the bleeding.
 - 8- In cases of severe coma, the patient needs to enter intensive care.
 - 9- If the patient improves, they may be evaluated for a liver transplant from a living donor.

Part Five

The Urinary System

The urinary system consists of the kidneys, ureters, bladder, and urethra.

The kidney contains about a million units of nephrons, each of which consists of glomeruli and tubules.

1. Functions of the Kidney:

- 1- Maintaining balance between the amount of fluids and salts in the blood plasma within the body, as well as between the acidity and alkalinity of the blood.
- 2- Disposing of waste products of protein metabolism.
- 3- Secretion of the hormone Erythropoietin and Vitamin D activation.
- 4- Regulating blood pressure.
- 5- Assisting in maintaining blood glucose level (Gluconeogenesis).

❖ Diagnosis of Kidney Diseases:

1- **Urine analysis:**

- A urine sample is taken from the mid-stream of the urine flow early in the morning.
Mid Stream Urine (MSU).
- Urine is examined chemically and microscopically:
 - The normal color of urine ranges from clear to dark yellow, and this change depends on the concentration of the pigment Urochrome in the urine.
 - The normal amount of urine ranges from 800 to 2500 cm³ per day.

A) Chemical urinalysis:

- 1- Urine is examined chemically using strips impregnated with detectors (dipstick) to estimate pH, albumin proteins, hemoglobin, glucose, ketones.

B) Microscopic Urine Examination:

Examination Results	Causes
1- Casts Or a mass formed in the kidney cavity.	
a) Red blood cell casts.	-Glomerulonephritis
b) White blood cell casts.	- (Acute Pyelonephritis)
c) Fatty casts.	- Nephrotic Syndrome
d) Granular casts.	- Non-Specific renal / Parenchymal disease
2- Cells.	
A) Red Blood Cells.	Urinary Tract Inflammation
b) White blood Cells.	Urinary Tract Inflammation
c) Eosinophils.	Interstitial nephritis
3- Crystals.	
a) Uric acid.	Gout - Acidic Urine
b) Calcium phosphate	Alkaline Urine
c) Calcium oxalate	Acidic Urine

2- Kidney Function Tests (Renal Function tests):

A) **Glomerular Function (GFR):** Glomerular Filtration Rate:

1. Measuring the level of **Creatinine** and **Blood Urea Nitrogen (BUN)** in the plasma. The creatinine level ranges from **0.8 - 1.3 mg/100 ml** in men and **0.6 - 1.1 mg/100 ml** in women.

And the ratio between creatinine and Blood Urea Nitrogen (BUN) level is 1:10.

If the creatinine level reaches the acceptable upper limit, which is 1.3 mg/dL in men, this means a loss of 50% of glomerular function.

2- **Creatinine Clearance:**

It is measured according to the following formula:

Creatinine Clearance = (Urine Creatinine × Urine Volume) / (Plasma Creatinine Level)

$$\text{Creatinine Clearance} = \frac{U \times V}{P}$$

U = Creatinine in urine mg/mL.

V = Urine volume mL/minute (24-hour urine volume / 1440).

P = Plasma creatinine level mg/mL. (The normal range is 95-105 mL/minute)

B) Tubular Function:

- 1- Maintaining the specific gravity of urine, ranging from 1015-1025.
- 2- Balancing the amount of fluids, salts, alkalinity and acidity of the blood within the body.
- 3- Urine pH is less than 5.3.
- 4- Fractional Excretion of Sodium (Fe Na).

It is the small portion of sodium salt that is filtered by the glomeruli and excreted in the urine.

It is measured according to the following formula:

(Fe Na)= $\frac{UNa}{PNa} \left(\frac{Pcr}{Ucr} \right)$

$$\frac{\text{Urinary Sodium (mEq/100 mL)}}{\text{Serum Sodium (mEq/100 mL)}} \times \frac{\text{Serum Creatinine (mg/mL)}}{\text{Urinary Creatinine (mg/mL)}}$$

This test is used to differentiate between renal failure resulting from loss of a large amount of fluids or blood (Pre-renal failure), and renal failure resulting from injury to the renal parenchyma where the value is less than 1% in the first type and more than 1% in the second type.

5- Excretion of less than 150 mg/day of proteins (without albumin) in the urine in a normal person.

- If the amount of albumin in the urine ranges from 30- 300 mg/day, this indicates partial albumin loss, but if the amount exceeds 3.5 g/day, this is what is called nephrotic syndrome (nephrotic S).

C) Examination of the urinary tract (or urinary system):

- By plain X-ray (PUT) or contrast radiography (I.V.P), ultrasound, and also CT scan

D) Renal isotope scan:

E) Renal Biopsy: Taking a kidney sample.

Acute Pyelonephritis

The ascent of gram-negative bacteria like E. Coli, as well as gram-positive bacteria from the ureter to the kidney, causes bacterial infection of the renal parenchyma.

- Symptoms:

- 1 - Sudden pain in one or both loins (Loin Pain) radiating to the iliac fossa or suprapubic area.
- 2 - Sharp rise in temperature reaching 40 degrees Celsius.
- 3 - Chills – vomiting.
- 4 - Pain when urinating.
- 5 - Frequent urination in small amounts.

- Examinations (Diagnosis):

- 1 – Urine analysis chemically and microscopically.
- 2 - Urine culture and sensitivity.
- 3 - Complete blood count (C.B.C).
- 4- Ultrasound on the abdomen and pelvis (Pelvi-abdominal ultrasonography).

- Treatment:

- Administer antibiotics according to urine culture and sensitivity.
- If unable to wait for the culture result, Ciprofloxacin 500 mg tablets every 12 hours for one week can be used.

Chronic Renal Failure

Uraemic Syndrome

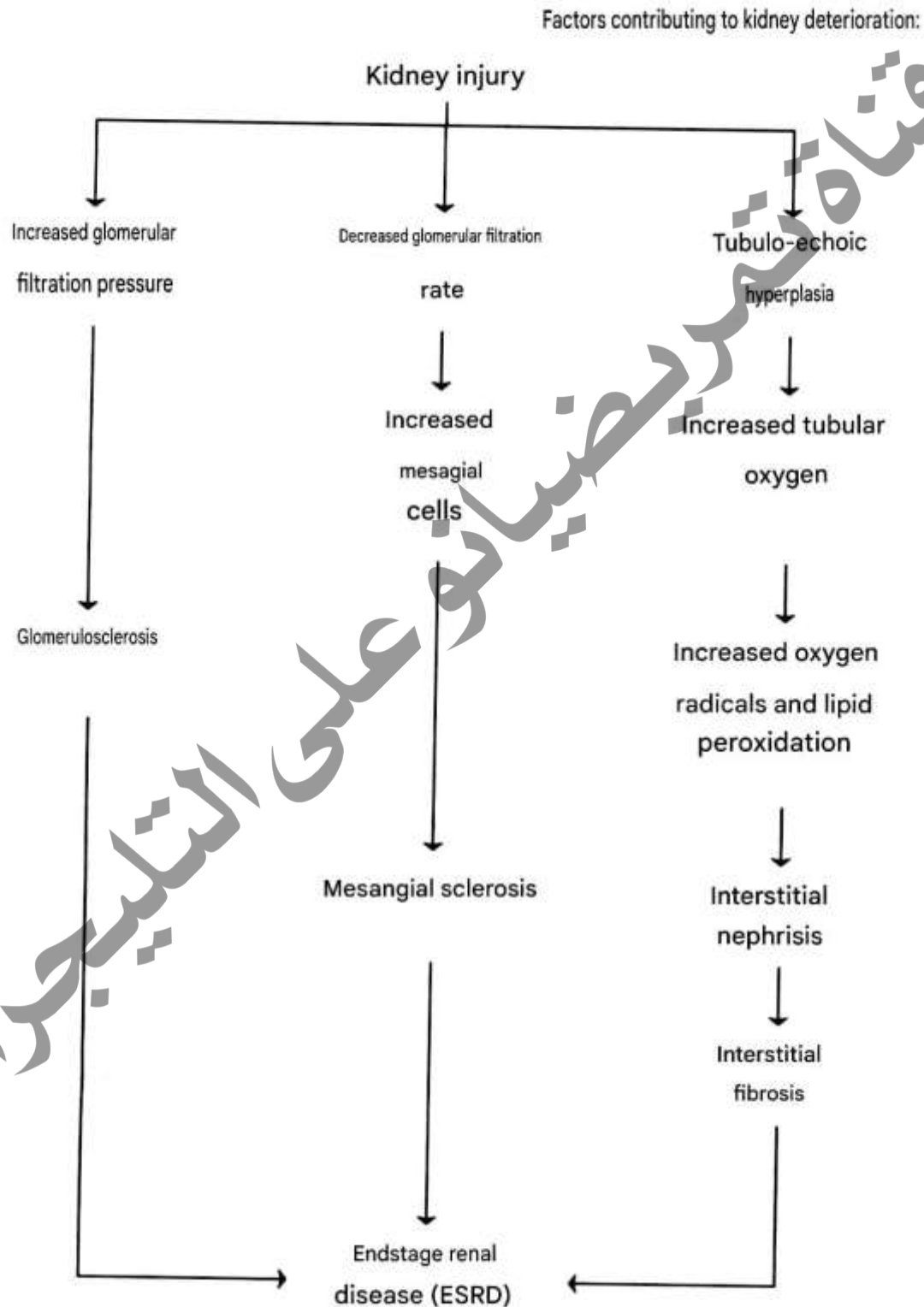
- **Definition:**

Chronic renal failure is an advanced, irreversible deterioration of kidney functions.

- **Causes:**

- Diabetes Mellitus (DM).
 - Hypertension.
 - Glomerulonephritis.
 - Hereditary diseases affecting the kidney such as Polycystic kidney disease.
 - Interstitial nephritis and chronic renal parenchymal inflammation (Chronic Pyelonephritis).
 - Autoimmune diseases affecting the kidney such as (SLE) Systemic Lupus Erythematosus.
 - Unknown cause.
 - Urinary tract obstruction (stones – ureteral stricture).
-

- Factors aiding kidney deterioration :-



- **Symptoms :**

- 1. Gastrointestinal System**

Loss of appetite, nausea, vomiting, gastric ulcer, gastrointestinal bleeding

- 2. Skin**

Itching of the skin, Yellowish discoloration of the skin

- 3. Muscles**

Muscle weakness, Gout

- 4. Nervous system**

Irritability, insomnia, loss of appetite, convulsions, coma

- 5. Peripheral nerves**

Tingling and numbness in the hands and feet

- 6. Blood**

Anemia, Bleeding, Impaired white blood cell function.

- 7. Endocrine System**

Menopause, Increased activity of parathyroid glands, Insulin resistance, Increased blood lipid levels.

- 8. Cardiovascular System and Heart**

Pericarditis, Increased susceptibility to atherosclerosis and weakness of heart muscle.

- 9. Respiratory System**

Pleurisy, Pneumonia, Pulmonary effusion

- 10. Fluid and Electrolytes**

High potassium level in blood (Hyperkalemia), Increased fluid volume in the body, Hyperphosphatemia, Hypermagnesemia, Acid-base imbalance.

❖ **Diagnosis and Treatment :-**

A- Diagnosis is made as previously studied :-

B- Identifying factors that can be treated such as :-

- 1- Sudden rise in blood pressure.
- 2- Urinary tract infection.
- 3- Urinary tract obstruction.
- 4- Blood loss due to bleeding or heart failure.
- 5- Septicemia.
- 6- Medications highly toxic to the kidneys like anti-rheumatics.
- 7- Diet: 40 gm protein per day.
80 gm carbohydrates.
90 gm fats.
- 8- Avoid foods rich in potassium like bananas.
- 9- Correction of serum electrolyte deficiency and regulation of fluid balance.
- 10- Renal replacement therapy.
 - a- Hemodialysis via an arteriovenous fistula in the arm, performed for 3-4 hours three times weekly.
 - b- Continuous ambulatory peritoneal dialysis.
Suitable for diabetic patients and the elderly with heart disease.
 - c- Renal Transplantation.

Part Six

Blood Diseases

Definition:

Blood is the thick red fluid and consists of 90% water, red blood cells (which transport oxygen - carbon dioxide), white blood cells (which defend the body against microbes), and platelets (responsible for blood clotting).

Table of normal Complete Blood Count (CBC) ranges

Hemoglobin percentage	Hb	15 gm%
Red Blood Cell count	RBC	5 million /mm ³
White Blood Cell count	WBC	4000-11.000 /mm ³
Platelet count	Platelets	140.000-400.000 /mm ³
Reticulocyte count	Reticulocytes	1 - 2 %
Erythrocyte Sedimentation Rate	ESR	5 - 20 mm/h

Anaemia

Definition :-

It is a decrease in the number of red blood cells or a decrease in hemoglobin below the normal range.
(Red cell count 4.5 - 5.5 million per cubic millimeter)

(Hemoglobin level 13-15 grams / dL)

❖ Causes:

1- Iron deficiency:

Lack of iron in the diet (meat - liver...) leads to symptoms of anemia.

Presence of continuous bleeding (such as esophageal varices - hemorrhoids).

Presence of hookworms (Ancylostoma).

2- Folic acid or Vitamin B12 deficiency.

Occurs in vegetarians or due to stomach diseases.

3- Deficiency of Vitamin C or E or thyroid hormone.

4- Bone marrow involvement (due to misuse of drugs or pathogenic viruses).

5- Some chronic diseases like tuberculosis - cancer - renal failure - liver cirrhosis.

6- Excessive destruction of red blood cells (Haemolysis) in hereditary diseases like Thalassemia - Sickle cell anemia - G6PD deficiency anemia.

❖ Symptoms of Anemia :-

1- Feeling fatigued with minimal exertion.

2- Memory weakness.

3- Depression - loss of appetite.

4- Angina - chest pain.

5- Difficulty breathing.

6- Increased heart rate (tachycardia).

7- Pallor of the skin and the oral mucous membrane.

8- Dizziness - blurred vision - continuous headache.

❖ **Treatment of Anemia**

- 1- Bed rest.
 - 2- Care for good nutrition orally.
 - 3- Administering vitamins – iron – proteins.
 - 4- Treating any disease accompanying anemia.
 - 5- Blood transfusion in severe cases.
-

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Blood Transfusion

❖ Indications for blood transfusion :-

- 1 Severe anemia.
- 2 Leukemia.
- 3 Internal and external bleeding.
- 4 Hemophilia patients.
- 5 Accidents - injuries - burns.

❖ Risks of blood transfusion :-

- Blood type incompatibility leading to acute hemolysis.
- Transmission of diseases like HCV - HBV - HCV - AIDS - HIV - Malaria - Syphilis.
- Fever - Allergies.
- Heart failure - pulmonary edema.
- Low calcium and occurrence of convulsions.
- Imbalance in potassium - sodium levels leading to heart arrhythmias.

❖ Precautions and Treatment :-

- Must carefully verify the patient's name as well as the patient's blood type and Rh factor on the blood bag.
 - Blood must be warmed very slowly before administration.
 - Take care not to shake the blood bag vigorously during carrying and transport.
 - The first 100 ml of blood is given under close observation for symptoms like (shortness of breath - chills - abdominal pain - body rash - red urine).
 - Immediately stop the transfusion and administer necessary treatment.
-

Leukemia

Definition :-

Leukemia is a large increase in the number of white blood cells due to a change in the bone marrow.

Symptoms :-

Occurs at any age but children are usually affected by acute lymphoblastic leukemia ALL and older adults are affected by chronic leukemias Chronic myelogenous leukemia (CML) - chronic lymphocytic leukemia (CLL).

Symptoms :-

- 1- Fever – sore throat – Body aches and bone pain.
- 2- Severe anemia – fatigue and exhaustion.
- 3- Enlargement of liver – spleen – lymph nodes.
- 4- Bleeding in various parts of the body such as under the skin – nose – gums..

Diagnosis of leukemia :-

- 1- Blood picture shows an increase in white blood cells and a decrease in red blood cells and platelets.
- 2- Performing a bone marrow examination.
- 3- Taking a sample of the liver – spleen – or lymph nodes.

Treatment of leukemia :-

- 1- Isolate the patient in a designated room.
- 2- Use gloves – masks – head and foot covers when dealing with the patient to prevent transmitting infection to the patient.
- 3- Blood transfusion or some of its products.
- 4- Administering chemotherapeutic drugs.
- 5- Using deep radiation therapy in some cases.
- 6- Performing a bone marrow transplant procedure.

Bleeding Tendency

It is considered one of the serious blood diseases that leads to increased bleeding and anemia.

Causes:

- 1- Deficiency in platelets, occurs in leukemia and bone marrow involvement.
- 2- Deficiency in clotting factors, occurs in some hereditary diseases like Hemophilia - liver diseases and Vitamin K deficiency.
- 3- Some blood vessel diseases.

Symptoms:

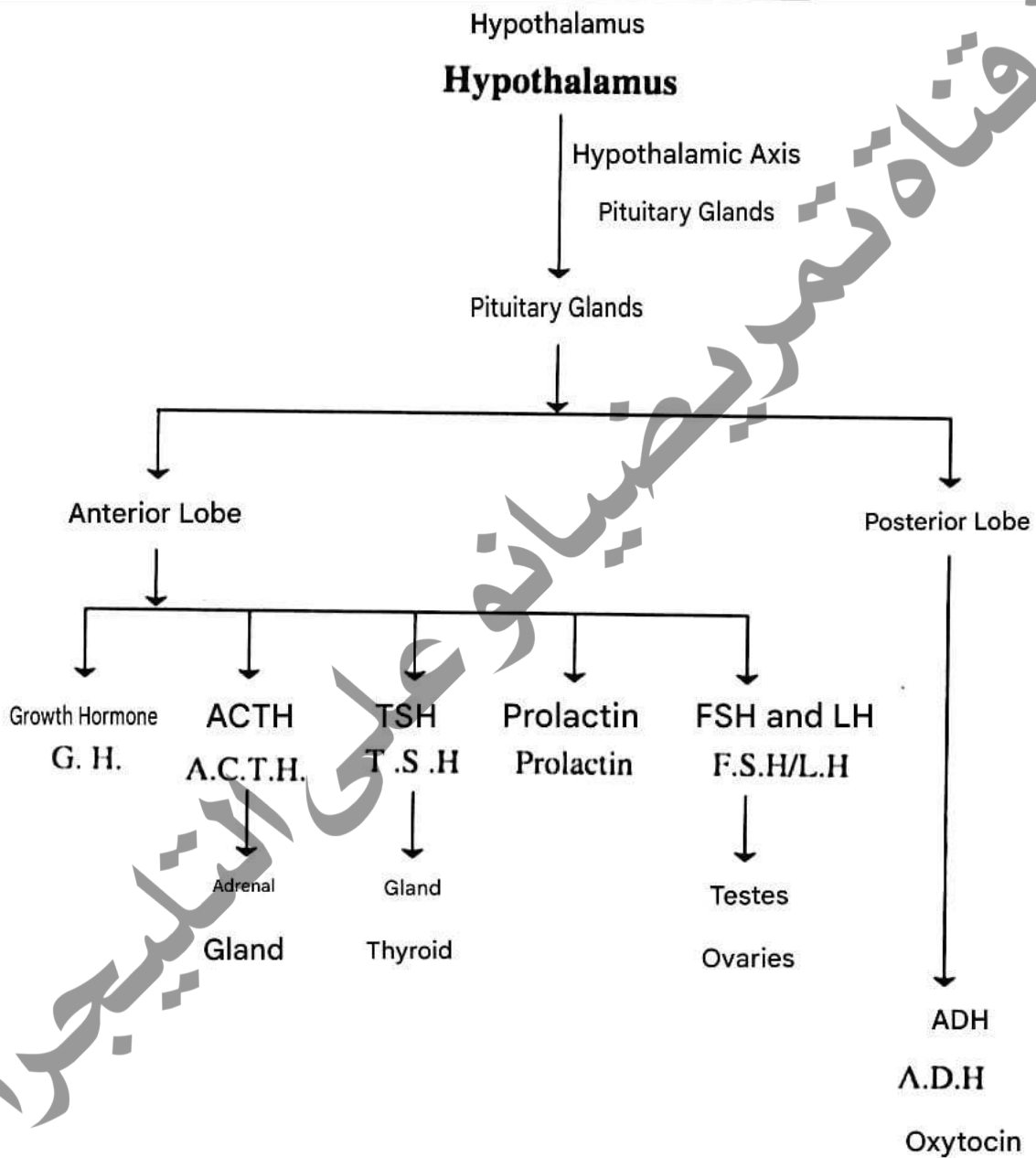
- 1- Bleeding from the nose – gums – hematemesis – rectal bleeding – urinary bleeding and others.
- 2- Bleeding under the skin (bruising).
- 3- Internal bleeding like brain hemorrhage – intestines.

Treatment:

- 1- Complete rest.
 - 2- Blood transfusion or plasma transfusion – or platelet transfusion.
 - 3- Vitamin K.
 - 4- Intramuscular or subcutaneous injections are prohibited, intravenous injection is preferred.
-

Part Seven

Endocrine Glands



Endocrine Glands:

The hypothalamus controls most endocrine glands by secreting hormones that reach the pituitary gland via the (hypothalamic – pituitary) axis, which in turn secretes hormones to stimulate the function of the thyroid gland, adrenal gland, and gonads (testes in males and ovaries in females) and also controls growth hormone and prolactin through the anterior lobe. The posterior lobe stores hormones (Oxytocin and also ADH) which are actually secreted by the hypothalamus.

Also, there are the parathyroid glands (Parathyroid gland) (responsible for regulating calcium and phosphorus levels in bones and blood) and the pancreas gland which secretes insulin and glucagon.

Thyroid gland diseases

Thyroid gland secretes hormones T4 (Thyroxine T4) and T3 (Tri-IodoThyronine T3) under the influence of TSH (Thyroid Stimulating Hormone) secreted from the pituitary gland (anterior lobe), which is itself secreted under the influence of TRH (Thyrotropin-Releasing Hormone) from the hypothalamus.

Thyroid gland examination :-

- Thyroid function tests are performed as follows:
 1. Measuring the level of TSH, Free T4, Free T3 in plasma.
 2. Ultrasound on the thyroid gland.
 3. Radioactive iodine scan of the thyroid gland and also Technetium.
Technetium -99m, I^{123}
 4. Measuring the level of antibodies against Thyroglobulin (Thyroglobulin), TSH receptor antibodies.
 5. Fine needle aspiration of the thyroid gland for pathological examination.

- **Hyperthyroidism (Increased thyroid secretion)**

- **Causes**

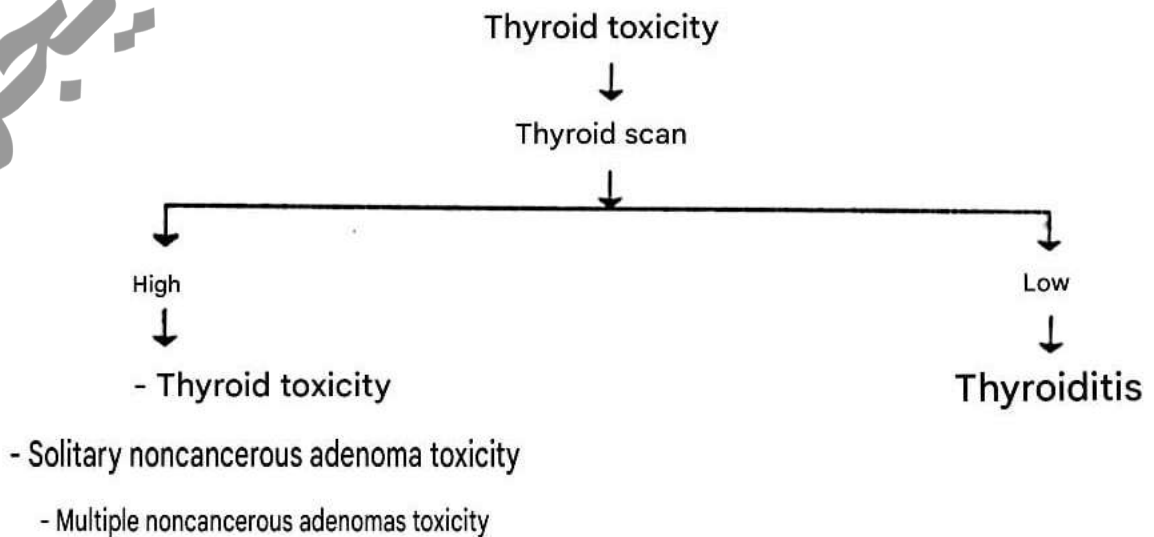
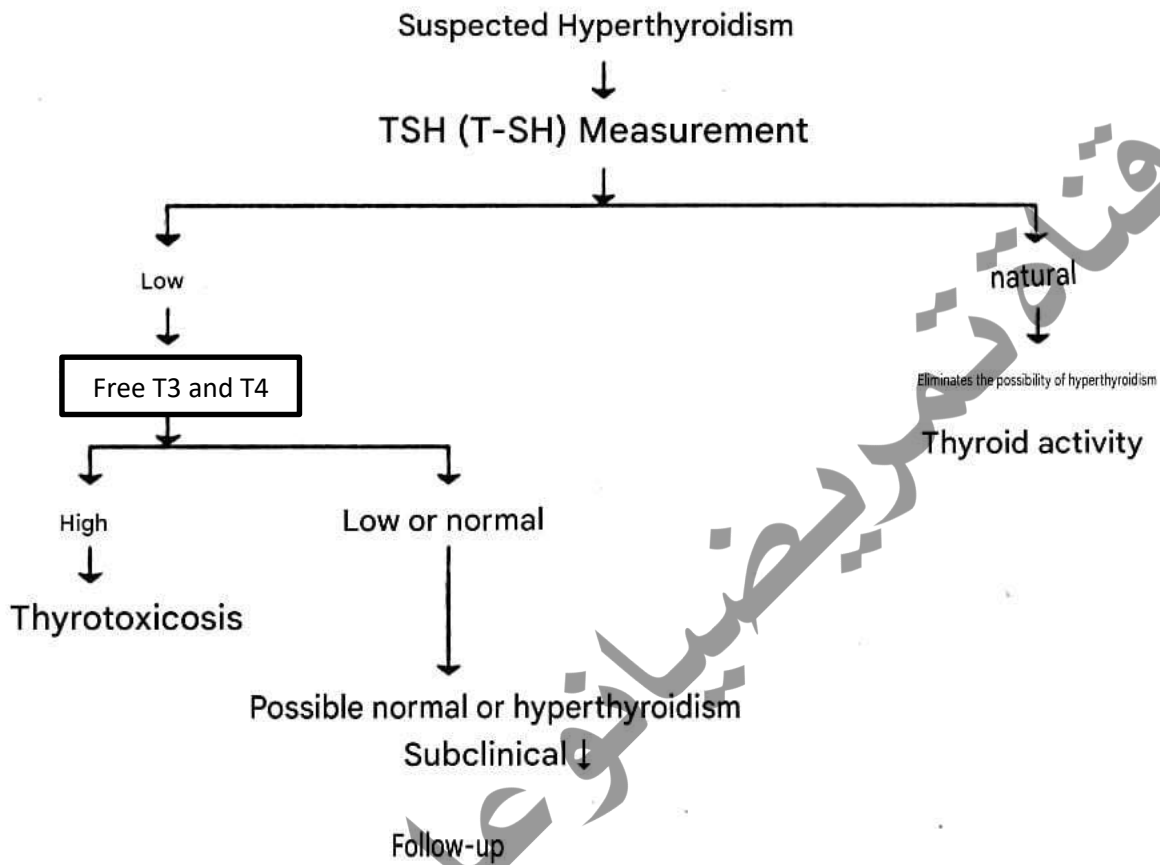
1. Graves' disease.
2. Toxic Solitary adenoma.

3. Toxic multinodular (non-cancerous)
 4. Thyroiditis.
 5. Thyroid tumors.
 6. Pituitary tumors.
-

Symptoms of Hyperthyroidism:

1. Nervousness / Anxiety.
 2. Excessive sweating.
 3. Palpitations.
 4. Fatigue, weight loss with increased appetite.
 5. Shortness of breath.
 6. Hand tremors.
 7. Exophthalmos (protrusion of the eyes).
-

- **Diagnosis:**



- **Treatment:**

1- Carbimazole tablets (Neomercazole) 5mg.

Week zero-3: 15 mg every 8 hours.

Weeks 4-8: 10 mg every 8 hours.

Then continue the dose at 5-20 mg daily for 12-18 months.

- **Side effects of the drug:** Decreased white blood cells - skin rash.

2- Beta-blockers (Propranolol) are added (Propranolol 160 mg/day).

3- Surgical treatment.

4- Radiotherapy.

Hypothyroidism (Hypothyroid)

- **Causes:**

- Autoimmune destruction of the thyroid gland
- Autoimmune thyroiditis
- Radiotherapy (radiation therapy)
- Surgical removal of the thyroid gland (thyroidectomy)
- Medical drugs such as lithium
- Hypothalamic disorders

Symptoms:

- **Children:** Delayed mental development - learning difficulties - delayed puberty.
 - **Adults:** Fatigue - cold intolerance - lethargy - weight gain - constipation - joint and muscle pain - menorrhagia (heavy periods) - hair loss - dry/rough skin - slow heart rate (bradycardia).
-

Diagnosis:

- Decreased level of free T4 and T3 in plasma.
 - Elevated TSH level.
 - Anemia.
 - Increased blood cholesterol level.
-

Treatment:

The patient is given Levothyroxine tablets 100 - 150 micrograms / day.

Elderly patients start with 25 micrograms / day and increase by 25 micrograms every two weeks.

Diabetes Mellitus (DM)

Definition of Diabetes Mellitus :-

Diabetes mellitus constitutes a varied group of metabolic diseases characterized by chronic elevation of blood sugar levels and disturbance in the metabolism of carbohydrates, fats, and also proteins, resulting from a deficiency in the secretion of insulin hormone or a deficiency in the hormone's effectiveness.

The chronic elevation of fasting and postprandial (2-hour) blood sugar levels is considered directly responsible for the acute and chronic complications of diabetes which subsequently affect all vital organs and systems of the body.

Classification of Diabetes Mellitus :-

Diabetes is classified according to the causes of its onset :-

1- Type 1 Diabetes (T1 DM):

Usually occurs in those under 30 years old but can occur in older adults; in this case, it is called Latent Autoimmune Diabetes of Adulthood (LADA).

Type 1 is characterized by an absolute deficiency of the hormone insulin due to autoimmune destruction of the pancreatic gland.

2- Type 2 Diabetes (T2 DM):

Usually occurs in those over 40 years old but is also diagnosed in patients as young as 9 years old who are characterized by extreme obesity and lack of physical activity; in this case, it is called Maturity-Onset Diabetes of the Young (MODY).

- Type 2 diabetes arises due to:
 - Insulin resistance.
 - Functional disorder of the beta cells responsible for insulin secretion.
 - Increased sugar absorption in the small intestine.
 - Increased sugar production from the liver.

3- Gestational Diabetes Mellitus (GDM):

It appears with a rise in blood sugar level during pregnancy, and it usually disappears after delivery.

4- Other types of diabetes that occur as a result of:

1. Diseases affecting the pancreas.
2. Increased secretion of growth hormone and corticosteroids.
3. Use of some drugs such as diuretics.
4. Genetic factors that cause reduced tissue sensitivity to insulin hormone.
5. Deficiency or defect in insulin receptors on the tissue level.

Diagnosis of Diabetes Mellitus:

(1) Main symptoms of Type I and Type II Diabetes:

Symptoms	Type I	Type II
Frequent urination + thirst	++	+
Feeling fatigued	++	+
Excessive eating + weight loss	++	—
Blurred vision	+	++
Genital itching in females	+	++
Numbness in extremities	+	++
Nocturnal enuresis	++	—

(2) Laboratory Examination:

Diabetes mellitus is diagnosed when fasting blood glucose level is higher than **126 mg/dL**, or blood glucose level two hours after eating (postprandial) is higher than **200 mg/dL**, or a random blood glucose level is higher than **200 mg/dL** in more than 1 reading.

Plasma Glucose Level	Normal	Impaired Fasting Glucose	Impaired Glucose Tolerance	Diabetes Mellitus
Fasting	< 110 mg/dL	≤110 - 125 mg/dL	–	≥ 126 mg/dL
2 hours postprandial	< 140 mg/dL	–	≤140 - 199 mg/dL	≥ 200 mg/dL
Random	–	–	–	≥ 200 mg/dL

Monitoring Diabetes :-

Diabetes is monitored as follows:

- 1- Measuring capillary blood sugar level by the patient themselves throughout the day.
- 2- Measuring fasting and 2-hour postprandial plasma glucose levels every 3 months.
- 3- Measuring the level of Glycated Hemoglobin (HbA1c) every 2 months (normal is less than 6%).
- 4- Measuring lipid levels and kidney function.

Diabetes Treatment :-

Goals of Diabetes Treatment :-

- 1- Treating the symptoms the patient complains of.
- 2- Expediting the return of the patient's physiological and social state to normal.
- 3- Preventing or delaying the complications associated with diabetes.

Treatment System :-

The treatment plan begins as follows :-

(A) Educational Therapy :-

- Simplistic explanation of the nature of the disease, its symptoms, and complications for the patient.
- Educating the patient on foot care.
- Teaching the patient to measure blood sugar levels using modern, available devices for immediate testing.
- The patient is advised to quit smoking and avoid nervous stress.
- The patient is advised to walk for half an hour daily.

(B) Nutritional Therapy:

Nutritional therapy differs from patient to patient according to dietary habits, physical effort, nature of work, and financial circumstances.

It is advised that the calories be balanced so that the percentage of proteins ranges from 15-20%, fats 30%, and carbohydrates 50-60%.

(C) Exercise:

Daily exercise for half an hour is recommended, particularly walking, due to its benefits for weight control, reducing lipid and cholesterol levels, and possibly reducing medication doses.

(D) Drug Therapy:

1- Oral hypoglycemic agents.

2- Insulin.

We use oral hypoglycemic agents or insulin to achieve the following target sugar levels.

	Normal	Target
Fasting Sugar	< 100 mg/dL	< 110 mg/dL
2-hour Postprandial Sugar	< 140 mg/dL	< 140 mg/dL
Pre-bedtime Sugar	< 110 mg/dL	100 - 140 mg/dL
HbA1c Level	< 6 %	< 6.5%

Insulin

- Insulin use is recommended for treating diabetes in the following cases:
 - Type 1 diabetes.
 - Type 2 diabetes unresponsive to treatment with oral hypoglycemic agents and dietary control.
 - During surgeries, infections, and during pregnancy.
 - When elevated blood sugar is accompanied by ketones in the blood and increased blood acidity (Diabetic Ketoacidosis).

The insulin dose is calculated as 0.5 to 1 unit of insulin per kg of body weight over the course of a day. The total calculated dose is divided as follows: 2/3 of the dose in the morning and 1/3 of the dose in the evening.

• Types of Insulin

Insulin Type	Onset of Action	Peak Action	Duration of Action
Regular (Actrapid, Humulin R)	1/2 – 1 hour	2 – 3 hours	5 – 8 hours
NPH (Insulatard, Humulin N)	2 – 4 hours	6 – 8 hours	6 – 16 hours
Lente (Novolente, Humulin L)	6 – 8 hours	8 – 12 hours	12 – 24 hours
Premixed (Humulin 70/30) (Mixtard 70/30)	4 – 8 hours	8 – 14 hours	25 – 36 hours
Ultra-Short acting Lispro (Humalog, Novorapid)	< 15 minutes	1 hour	4 – 5 hours
Ultra Lente Glargine (Lantus)	1 – 2 hours	–	22 – 24 hours

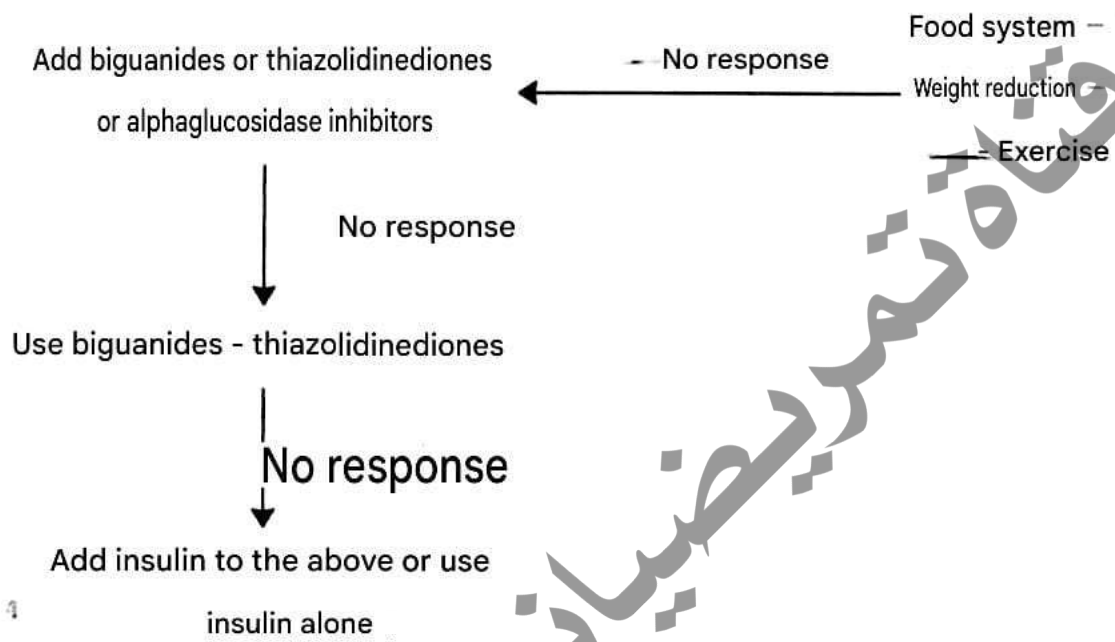
Oral Hypoglycemic Agents :-

They are used to treat Type 2 diabetes and are divided into several types :-

Type	Sulfonylurea (e.g., Diamicron)	Biguanides (e.g., Glucophage)	Alpha- Glucosidase Inhibitors (e.g., Acarbose)	Thiazolidinediones (e.g., Avandia)	Meglitinides (e.g., Novonorm)
Mechanism	Increases insulin secretion from pancreas	Reduces hepatic glucose production & improves insulin resistance	Reduces sugar absorption from food in small intestine	-Improves insulin resistance -Reduce hepatic glucose production	Increases insulin secretion from pancreas
Patient Profile	Age > 30 years - Normal weight	-Obese - High fasting sugar - High lipids	High 2-hour postprandial sugar	-Obese - Impaired kidney function - High lipids	High 2-hour postprandial sugar
Timing	30 mins before meals	With meals	With meals	With breakfast	15 mins before meals

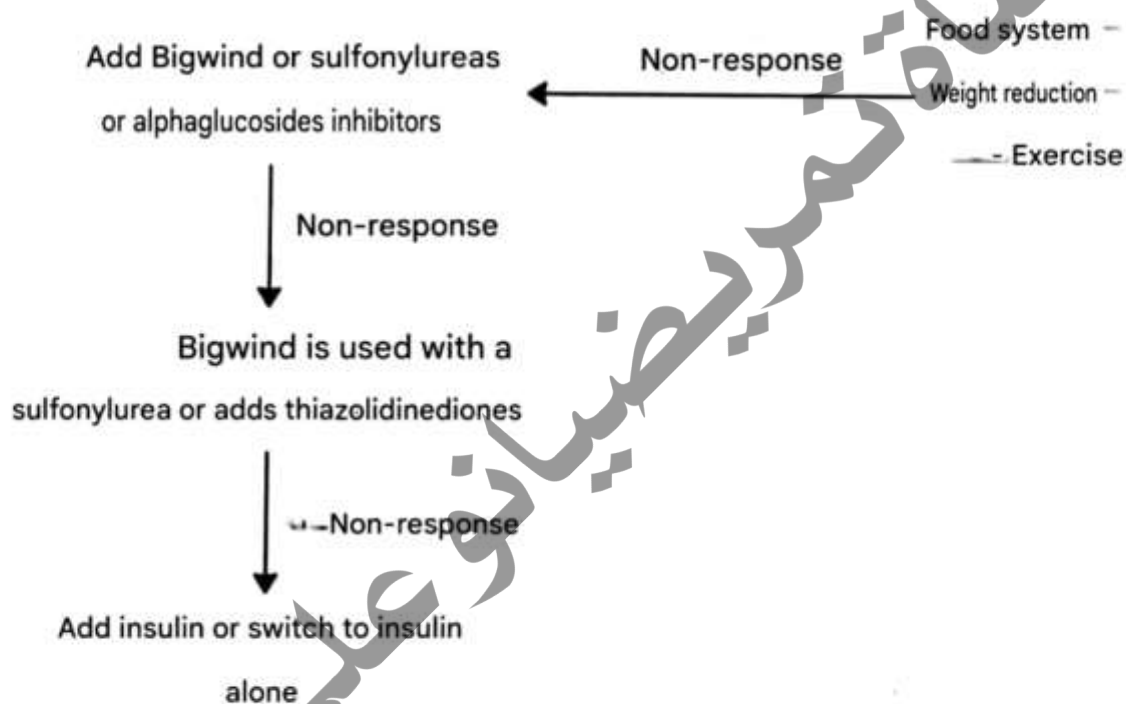
Illustrative diagram for treating Type 2 diabetic patients with obesity :-

An illustration of the treatment of type 2 diabetes patients with obesity:



Illustrative diagram for treating Type 2 diabetic patients with normal weight :-

Illustration of the treatment of type 2 diabetes patients with normal weight:



Part Eight

Infectious Diseases

Infectious diseases can affect all organs or different body systems. Infectious diseases differ from other diseases in that they are caused by microscopic living organisms that include bacteria – viruses – fungi – and parasites. Infection is transmitted from human to human via multiple routes.

Source and Route of Infection	Method of Transmission	Examples of Infectious Diseases
A- Contact :- 1- Human to human contact: 2- Contact with soil: 3- Exposure to contaminated water.	Skin and mucous membrane contact Wounds - Abrasions Skin penetration	-Scabies -Tetanus -Schistosomiasis
B- Airborne :-	Droplets - Dust	Measles – Rubella (German Measles) - Meningitis
C- Fecal-Oral route:-	Contamination of food/water with fecal waste	Typhoid - Hepatitis (A)
D- Through Placenta:-	Mother's blood	Rubella (German Measles) - Toxoplasmosis
E- Medical intervention :-	Syringes	Viral Hepatitis (B)
F- Animals :-	Eggs Milk	→ Brucellosis Tuberculosis

Prevention of Infectious Diseases :-

- Personal hygiene.
- Ensuring that food and drinking water are free from contamination
- Vaccination against infectious diseases

Typhoid Fever
Salmonella Typhi
Salmonella Paratyphi

Humans get typhoid fever by consuming food or drink contaminated with the feces of a carrier of the causative bacteria or a patient infected with typhoid. For infection to occur, ingestion of approximately 10^6 to 10^8 bacteria is necessary.

Salmonella penetrates the intestinal wall into the lymphatic follicles (Peyer's Patches), where it multiplies and then enters the blood, then to the liver, spleen, bone marrow, and also the gallbladder where it leads to chronic carriage and the patient becomes a carrier of the disease.

The incubation period of the disease is 14 days.

Symptoms :-

First Week: Gradual rise in temperature for 4-5 days - chills - headache - vertigo - pain in the limbs - constipation or diarrhea - pulse is slow relative to the fever (Relative Bradycardia).

End of the First Week: Rose spots appear on the upper abdomen and back that fade upon pressure - enlarged spleen - abdominal distension.

End of the Second Week: Complications may occur such as:

- Loss of concentration - Peritonitis - Osteomyelitis.

Diagnosis :-

- 1 - Complete blood count (presence of decreased white blood cell count - Leukopenia).
- 2 - Blood culture in the first week.
- 3 - Stool culture in the second week.
- 4 - Widal test to confirm the presence of antibodies against Salmonella bacteria in the third week with a steady rise, but it is an unreliable test.

Treatment :-

Ciprofloxacin tablets 500mg every 12 hours for 14 days.

Or Cefotaxime IV 1 gram every 12 hours for 14 days.

Treatment of the carrier :-

Ciprofloxacin tablets 500mg every 12 hours for 4 weeks, and if there's no response, cholecystectomy is considered.

Prevention of Typhoid :-

- Vaccination of contacts with the protective vaccine.
- Use of pure water.
- Ensuring the health of food handlers in commercial establishments and hotels.

Measles

Infection occurs through inhalation of droplets from a measles patient carrying the (Paramyxovirus). It often affects children after the disappearance of antibodies transferred to them from the mother via the placenta. The incubation period is 10 days. The infection can be transmitted from the patient to contacts from the start of symptoms until the disappearance of the skin rash.

Symptoms :-

Day 1: Fever - runny nose - increased eye secretions.

Day 2: Cough - patient's fear of light (Photophobia) - small white spots on the buccal mucosa (Koplik's Spots).

After three or four days: Red skin rash behind the ears and on the forehead spreads within hours all over the body.

Day 6 or 7: Fever drops and the skin rash begins to disappear.

Complications :-

- Otitis Media – Pneumonia
- Encephalitis – Cachexia - weight loss.

Treatment :-

Isolate the patient for 10 days from the onset of the rash.

Prevention :-

Vaccinate children with the (Measles - Mumps - Rubella) MMR vaccine subcutaneously.

- Administer Immunoglobulin to contacts of measles patients under 18 month old and immunocompromised children.

German Measles (Rubella)

Infection is transmitted by inhaling droplets from a patient carrying the (Toga Virus).

The incubation period is 18 days. It usually affects older children and adolescents.

When the disease occurs in adults, its severity increases and it can affect pregnant women in the first four months of pregnancy.

Symptoms :-

- 1- Slight fever.
 - 2- Skin rash on the face and neck spreads to the body then to limbs.
 - 3- Enlarged lymph nodes in the neck.
 - 4- Joint pain and decreased platelets in adults.
-

Treatment :-

- Bed rest and patient isolation.
 - In case a pregnant woman contracts the disease during the first four months of pregnancy, termination of pregnancy is recommended to prevent fetal malformations.
-

Prevention :-

- Administer MMR vaccine.

Mumps

Infection is transmitted by inhaling droplets from a patient carrying the virus.

The incubation period is 18 days.

- **Symptoms:**

- 1- Fever.
- 2- Pain at the angle of the jaw followed by painful swelling of the salivary gland (parotid).
- 3- Swelling of other salivary glands (Submandibular gland).

- **Complications:**

- 1- Orchitis (Testicular inflammation).
- 2- Oophoritis (Ovarian inflammation).
- 3- Pancreatitis.
- 4- Meningoencephalitis (Inflammation of the brain and meninges).

- **Treatment:**

Patient isolation + symptomatic treatment.

Use of corticosteroid drugs if orchitis occurs.

- **Prevention:**

Use of MMR vaccine.

Part Nine

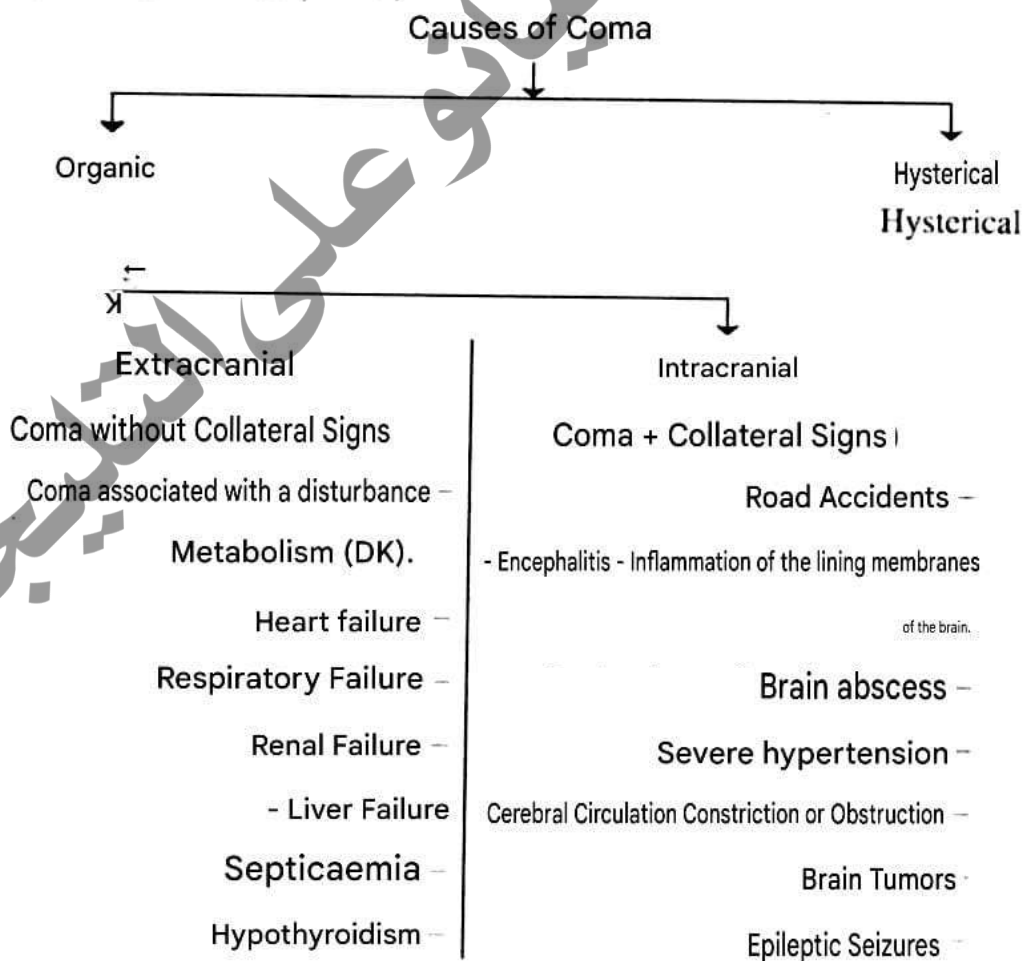
Nervous System Diseases

Coma

The state of wakefulness depends on :-

- Integrity of the Reticular Activating, which is a group of nuclei in the brain stem, hypothalamus, and thalamus. It receives stimuli and alerts and transmits them via a set of ascending nerve fibers to the cerebral cortex to activate it for performing vital functions.
- Integrity of the Cerebral Cortex.

Causes :-



Lateralizing neurological signs of cerebral coma :-

- 1- Facial asymmetry.
- 2- Asymmetric deep tendon reflexes.
- 3- Deviation of the eyes to one side.
- 4- Head turning to a specific side.
- 5- Decreased or increased muscle tone on one side (Hypotonia or Hypertonia).
- 6- Unequal pupils.

Diagnosis :-

- 1- Obtaining medical history from relatives.
- 2- Measuring blood pressure, heart rate, and temperature.
- 3- Observing respiratory rate.
- 4- Observing blood glucose level.
- 5- Smell of the breath: Acetone smell (coma resulting from high blood sugar), Ammonia smell (hepatic coma).
- 6- Skin texture (dry – moist – cracks – injection marks...).
- 7- Neurological examination.
- 8- Complete blood count.
- 9- Liver and kidney functions.
- 10- Lumbar puncture (CSF analysis).
- 11- CT scan of the brain – MRI Magnetic Resonance Imaging.

Treatment :-

- 1- Treat the cause.
- 2- Care for the comatose patient.
Turn the patient every two hours, clean the skin with alcohol, especially the back and pressure points.
- 3- Suction of respiratory secretions.
- 4- Insert a urinary catheter.
- 5- Calculate fluid input and output.
- 6- Daily enema.
- 7- Insert a nasogastric tube for blenderized food.
- 8- Insertion of an intravenous catheter to administer the required fluids.

Meningitis

Meningitis is inflammation of the membranes surrounding the brain and occurs as a result of a microbial infection (bacteria – viruses – fungi) or non-microbial causes.

The causative bacteria are:

- Neisseria meningitidis
- Haemophilus Influenzae
- Pneumococcus
- Streptococcus

The disease appears as sporadic cases or in epidemic form.

The disease is transmitted via droplets, especially in crowded places like schools and military barracks. It usually affects people under 15 years of age.

The bacteria live in the nasopharyngeal area (Nasopharynx) and reach the brain membranes via the blood or directly by spreading from a nearby focus like the sinuses.

Symptoms

Fever – irritability – confusion – deterioration in concentration and consciousness – neck muscle stiffness.

Skin rash characterized by a purplish color (Petechiae/Purpura).

Some patients (20% of patients) suffer from seizures.

Diagnosis

(Lumber Puncture)

1- Lumbar puncture and examination of cerebrospinal fluid.

A- Increase in the number of Neutrophils (500- 10,000 cell/ mcL)

B- Decreased glucose level less than 40 mg/dL (Normal ~50-80 mg/dL).

C- Increased protein level more than 150 mg/dL (Normal ~20-40 mg/dL*).

D- Increased cerebrospinal fluid pressure.

2- Identifying the type of causative bacteria by Gram stain.

3- Culture of cerebrospinal fluid, blood, fluids from skin lesions, and also a nasopharyngeal swab.

Treatment:

Upon confirming infection with *Neisseria* or *Pneumococcus* (*Pneumococcus*), give Ceftriaxone 2 grams IV / every 12 hours.

Or aqueous penicillin 20 million units IV daily.

In case of *Haemophilus influenzae* bacteria, the patient is given Ceftriaxone 2 grams / every 12 hours.

Prevention:

- Contacts are given Rifampicin 600 mg tablets for 2 days (adults).
 - Children under 1 year: 5 mg/kg/12 hours Rifampicin.
 - Children are given 0.5 ml subcutaneously of the vaccine against *Neisseria* bacterial infection. (vaccination)
-

Part Ten

Dermatological and Venereal Diseases

Study Objectives :-

- 1- Recognizing some of the most common skin diseases.
 - 2- Teaching the methods of transmission of infectious skin diseases and methods of prevention.
 - 3- Recognizing different treatment methods with a focus on the role of nursing in treatment and dressing.
 - 4- Explaining the symptoms of skin diseases accompanying some internal diseases.
-

Some Common Infectious Skin Diseases

Skin diseases resulting from parasitic infections

1- Scabies:

- A contagious disease caused by the scabies mite.
- Incubation period 14 days.
- Symptoms: Severe itching, especially before sleep, and affects more than one family member.
- The disease spreads through contact.
- The disease appears as small papules, especially between the fingers, as well as the legs, arms, armpits, thighs, genitals, and under the breasts in women.
- It is noted that the disease does not affect the face or upper back.

Treatment:

- The patient should be isolated, and their bedding and tools should not be used; patient's clothes must be washed separately.
- If the patient is hospitalized, he must be isolated and the hospital administration notified.
- Treat the patient and all family members simultaneously due to the possibility of them being in the incubation period.
- Give the patient antihistamines to treat itching.

- Topical treatment for scabies: Bath with warm water and soap, scrubbing the body with a loofah, in the evening before bed. After drying the body, apply one of the specific scabies ointments like Benzyl Benzoate, Lindane (Gamene), or Permethrin (Actomethrin) to the whole body except the head. Repeat the treatment for three consecutive days.

2- Pediculosis (Head Lice):

- A parasitic disease affecting the scalp and hair.
- This disease affects school children, especially girls, and causes severe itching with the appearance of pimples and boils on the head accompanied by swollen lymph nodes in the neck.

Treatment :-

Hair must be washed well; shaving is not necessary. Comb the hair thoroughly with a fine-toothed comb and apply anti-lice creams.

Skin diseases resulting from viral infections

1- Warts :-

- Appear as small papules on the skin surface.
- The disease is transmitted through touch and does not cause itching.
- The disease spreads to many body areas like the face, hands, and feet.
- There is a type of wart transmitted through sexual contact that affects the genitals.

Treatment :-

- Warts can be removed by electrocautery, cryotherapy (freezing), or laser.
- Warts can also be treated topically with salicylic acid.
- Treatment of genital warts with Podophyllin solution; this application must be done by a doctor and washed off after four to six hours.

2- Chicken Pox:

- A viral disease transmitted via droplets that affects children.
- Incubation period: two weeks.
- The disease appears as small papules, then small vesicles appear on them, which then open and crust over.
- This is accompanied by a slight fever.

Treatment :-

The disease resolves spontaneously after two weeks, but the skin must be cared for with antiseptics, and the crusts should not be removed from the skin surface to avoid scarring.

3- Herpes Simplex:

- The disease appears as small yellow vesicles accompanied by skin redness.
- The disease affects areas near skin openings like the nose and mouth.
- Another type is transmitted through sexual contact and affects the genitals.

Treatment :-

- Antiviral drugs must be given orally (such as Acyclovir or Zovirax).
 - Topical treatment: Antiseptic like alcohol or Gentian Violet (blue solution).
-

4- Herpes Zoster

- **Shingles**
- The disease appears as small yellow vesicles accompanied by skin redness and affects a specific area of the body like a nerve dermatome (e.g., arm or leg) and may appear on the back or chest on one side of the body or on the face.

- The disease is accompanied by severe pain at the site of the vesicles, and the pain may persist after they disappear (Postherpetic Neuralgia).

Treatment :-

- Pain relievers should be given.
- Antiviral drugs must be given orally (such as Acyclovir or Zovirax).
- Skin care with antiseptics.

Skin diseases resulting from bacterial infections

- Healthy skin is not penetrated by bacteria, so for bacterial skin diseases to occur, there must be a defect in the skin, such as the itching associated with scabies and eczema, or internal diseases like diabetes or kidney diseases.
- The disease appears as small pustules that enlarge and become raised above the skin surface, fill with pus, and become boils (furuncles), and it is possible for them to spread to other areas of the body.

Treatment :-

- Treat diseases that cause skin itching.
- Administer oral antibiotics.
- Treat skin lesions with an antiseptic like Potassium Permanganate solution 1:8000 and apply an antibiotic ointment (e.g., Gentamicin ointment or Fusidic acid ointment).

Skin diseases resulting from fungal infections

- **Fungi infect the scalp, skin surface, or nails.**
- Fungi that infect the scalp cause Tinea Capitis (Ringworm).
- There are two types of Tinea Capitis.

A) Tinea Capitis (English Ringworm) :-

It is divided into three types:

1- Microsporic Type (Ectothrix) :-

- It is the most common type and affects the scalp in children.
- It appears as one or more defined areas on the head, round or oval in shape, covered with small greyish scales.
- The hair infected with the fungus breaks off, leaving it 1 to 3 millimeters long above the skin surface.

2- Black Dot Type (Endothrix) :-

- Affects children and adults and appears as a defined area on the head with slight inflammation around the hair follicles.
- The hair breaks off at the level of its exit from the scalp, so the affected area appears as a smooth spot with black dots.

3- Favus (Kerion) Type :-

- Affects children and adults and causes severe inflammation of the hair follicles.
- It results in crusts (scutula) or abscesses.
- This type destroys the hair follicles, and if treatment is neglected, it leads to hair loss and leaves scars where hair does not grow afterwards.

Treatment :-

- Isolate the patient and provide preventive care for contacts.
- Administer Griseofulvin tablets to the patient (the patient is given one tablet per 10 kg of body weight, and the medication must be given directly after meals).
- Apply an antifungal ointment to the scalp (e.g., Miconazole (Daktarin) or Ticonazole (Trosyd)).

B - Favus (Tinea Favosa) :-

- Affects children and adults of both sexes and is characterized by the appearance of dry, brittle, yellow-colored crusts with raised edges and a depressed center like a saucer, with a distinctive odor.
- The hair color changes and becomes dusty, but it does not break off.
- If treatment is neglected, it leads to destruction of hair follicles and leaves scars, and hair does not grow in the affected areas.

Treatment :-

- Similar to Tinea Capitis treatment, in addition to removing the yellow crusts using Salicylic acid ointment.

Fungi that infect the skin surface

Cause different types of Tinea (Ringworm) depending on the site of skin infection.

1- Tinea Corporis (Ringworm of the Body) :-

Appears as a defined red patch slightly raised above the skin surface, with vesicles or fine scales at its border.

Treatment :-

- Griseofulvin tablets.
 - Apply antifungal ointment.
-

2 - Tinea Cruris (Jock Itch) :-

Appears as skin inflammation on the inner aspects of the upper thighs, with a defined border slightly raised above the skin surface and covered with small vesicles or fine scales.

Treatment :-

- Care for cleanliness of underwear and personal hygiene.
 - Administer Griseofulvin tablets to the patient.
 - Apply antifungal ointment.
-

3 - Tinea Pedis (Athlete's Foot) :-

Inflammation of the skin between the toes accompanied by itching, death of the upper skin layers making them white and moist, and may be accompanied by a foul odor in the feet.

Treatment :-

- Keep the feet as dry as possible, as heat and moisture promote fungal growth.
 - Disinfect socks thoroughly.
 - Griseofulvin tablets.
 - Apply antifungal ointment.
-

4 - Tinea Unguium (Onychomycosis) :-

Affects nail growth, leading to roughness, pitting, and ridges, and loss of shine.

Treatment :-

- Griseofulvin tablets for a long period, or Fluconazole capsules (Diflucan), or Terbinafine (Lamisil) tablets.
-

5 - Tinea Versicolor (Pityriasis Versicolor) :-

- Characterized by the appearance of spots covered with scales ranging in color from yellow to dark brown.
- Appears on the chest, back, and arms. This type of superficial fungal infection is not accompanied by itching or inflammation.

Treatment :-

- Antifungal tablets (other than Griseofulvin) like Fluconazole, Itraconazole (Sporanox), or Ketoconazole.
 - Apply antifungal ointment.
 - Ultraviolet light therapy in some cases.
-

Some Non-Infectious Skin Diseases

1 - Psoriasis :-

- It is a non-contagious skin disease.
- Affects 1-2% of people and its cause is unknown.
- The disease appears as small red papules covered with a white scaly layer.
- When the white scale is removed, small bleeding points appear (Auspitz sign).
- The disease can appear anywhere on the body, but the most commonly affected areas are the knees, elbows, and lower back.
- It also affects the scalp and nails.

Treatment :-

- Topical treatment with Salicylic acid ointment and Cortisone ointment.
- Oral treatment with Methotrexate tablets.
- Phototherapy with ultraviolet light.

2 - Vitiligo:

- It is a non-contagious skin disease with an unknown cause.
- The disease appears as white patches in various parts of the body and does not cause itching.
- It can spread to large areas of the body.

Treatment:

- Topical application with Psoralen solution (e.g., Meladinine).
- Application with Cortisone ointment.
- Administering Psoralen tablets (e.g., Meladinine) followed by exposure to sunlight or ultraviolet light (PUVA therapy).

3 - Bullous Skin Diseases (e.g., Pemphigus, Pemphigoid):

This is a group of diseases that affect the skin, appearing as small vesicles and blisters containing fluid. When these lesions open, they leave ulcers on the skin.

Treatment:

- Administering high doses of Cortisone orally to the patient. This may cause hypertension and diabetes mellitus, **highlighting the nurse's role** in regularly monitoring blood pressure and testing blood and urine sugar for early detection of diabetes.
- **The nurse also plays an important role** in skin care for the patient, treating it like burned skin.

4 - Acne Vulgaris:

- A disease affecting adolescents of both sexes after puberty due to increased secretion of androgens (male hormones), which causes increased sebaceous secretions in the skin.
- The disease appears as pimples and pustules on the face, chest, and shoulders, and when they disappear, they leave scars on the skin.

Treatment:

- Reduce fatty foods in the diet.
- Topical application to the skin to reduce sebaceous secretions (e.g., retinoids, benzoyl peroxide).
- Topical antibiotic application.
- Oral antibiotics.

5 - Urticaria (Hives) :-

It is a non-contagious skin disease resulting from sensitivity to a group of substances such as food, medications, sun exposure, climate changes, or due to underlying bodily diseases.

The disease appears as wheals of varying sizes in different parts of the body, accompanied by severe itching, and sometimes the allergy can be severe enough to cause suffocation (anaphylaxis).

Treatment:

- In cases of acute urticaria, the patient must be treated immediately to prevent suffocation. The patient is given a subcutaneous Adrenaline injection or an intramuscular Cortisone injection.

- Thereafter, treatment is similar to chronic urticaria: oral antihistamines and calamine lotion to reduce the sensation of itching.
 - Identify the cause and avoid it.
-

6 - Papular Urticaria (Prurigo Simplex) :-

A type of allergy that affects children due to insect and mosquito bites.

- The disease appears as papules and small vesicles accompanied by severe itching that leads to bacterial infection appearing as small pustules.

Treatment :-

- Prevention of insect and mosquito bites.
 - Administer oral antihistamines.
 - Apply topical treatment with calamine or cortisone ointment.
 - Administer antibiotics if there is a bacterial infection.
-

Eczema (Atopic Dermatitis) :-

It is a type of skin allergy with a hereditary factor that affects various ages.

Eczema affecting children from 2 months to 2 years old :-

- Lesions appear on the face, especially the cheeks.
- It begins as redness of the skin covered with scales, then small vesicles appear on it.

Eczema affecting children from 5 years to 12 years old :-

- Affects the skin behind the knees, behind the elbows, and the neck.
- Appears as small papules accompanied by lichenification (thickening) of the skin.

Eczema affecting adults :-

- Lesions appear in the same areas as the previous type, accompanied by severe itching with increased skin lichenification.

Treatment :-

- Administer oral antihistamines.
 - Apply cortisone ointment.
-

Alopecia Areata :-

- A non-contagious disease leading to hair loss.
- There is a type that affects a specific part of the hair and appears as one or more hairless patches without scales or inflammation.
- Another type of disease affects all the hair on the head (Alopecia Totalis), and sometimes affects all body hair (Alopecia Universalis).

Treatment :-

- If the involvement is limited to a small area of the body, apply cortisone ointment or cantharides ointment.
 - In cases affecting a large part of the scalp or body hair, administer cortisone tablets.
-

Hirsutism in Women :-

- It is the appearance of hair in areas where hair does not usually appear in women, such as around the mouth, chin, cheeks, and neck.
- This causes distress and psychological disorders in women.
- This hair appears after puberty due to increased secretion of androgens (male hormones).

Causes :-

- The cause may be unknown (Idiopathic).
- Hereditary cause.

- Result of taking certain medications.
- Result of polycystic ovary syndrome (PCOS) leading to increased androgen secretion.
- Increased prolactin hormone secretion (Hyperprolactinemia).

Treatment :-

- Identify and treat the cause.
- Administer anti-androgen medications.
- Remove excess hair using laser devices or intense pulsed light (IPL) devices.

Skin Symptoms Associated **With Some Internal Diseases**

1- Renal Failure :-

- Renal failure is accompanied by itching in many parts of the body, and sometimes the itching is severe.

Treatment:

Administer oral antihistamines, and in some cases ultraviolet light therapy is used.

2- Diabetes Mellitus :-

Diabetes is accompanied by many infectious skin diseases, whether bacterial, fungal, or viral. In this case, diabetes must be treated, and the various infectious diseases must be treated with appropriate therapy.

3- Viral Hepatitis C :-

This disease is widespread in Egypt, and the appearance of many associated skin diseases has been observed, such as severe itching and urticaria, as well as the appearance of lichen planus, which affects the skin as violet papules on the surface of the skin, and presents as white streaks on the buccal mucosa inside the mouth..

Venereal Diseases

1- Syphilis :-

A serious venereal disease caused by infection with a spirochete bacterium called *Treponema pallidum*. Syphilis is divided into two types: Acquired Syphilis and Congenital Syphilis.

(A) Acquired Syphilis :-

- Infection is transmitted through sexual contact, contact with lesions, or blood transfusion.

- Acquired syphilis is divided into four stages :-
 - Primary Stage: Called the chancre (hard ulcer).
 - Secondary Stage: Characterized by the appearance of a rash on the skin, mucous membranes, and other organs of the body.
 - Tertiary Stage: Affects the skin, mucous membranes, bones, joints, and internal organs.
 - Quaternary Stage: Divided into cardiovascular syphilis and neurosyphilis.

(B) Congenital Syphilis:

- Infection is transmitted from mother to fetus via the placenta.
- Congenital syphilis is divided into :-
 - Early Congenital Syphilis: Occurs in the first two years of life and resembles secondary stage of acquired syphilis.
 - Late Congenital Syphilis: Symptoms begin to appear around the fourth and fifth years, characterized by the appearance of gummatous lesions as in the tertiary stage of acquired syphilis.

Treatment :-

Administer Penicillin injections.

2- Gonorrhea :-

A disease characterized by purulent discharge from the genitals, caused by infection with the bacterium *Neisseria gonorrhoeae*. The incubation period of the disease is from 2 to 7 days.

Modes of Transmission :-

- Sexual contact.
- Infection is transmitted from mother to child during childbirth if the mother is infected.

Symptoms of Acute Gonorrhea in Men :-

- Purulent discharge from the urethra and redness and inflammation of the urethral meatus.
- Burning sensation during urination with increased frequency of urination.
- Slight elevation in temperature.

Symptoms of Acute Gonorrhea in Women:

- 1- Purulent discharge from the urethra and vagina due to cervicitis.
- 2- Burning sensation during urination with increased frequency and presence of pus in the urine.
- 3- Discharge of pus from the cervical opening.

Treatment :-

- Avoidance of sexual contact.
 - Administer antibiotics.
-

3- AIDS (Acquired Immunodeficiency Syndrome) :-

It is a viral disease transmitted through sexual contact or blood transfusion and causes loss of the body's immunity.

Symptoms :-

- Weight loss.
- Chronic diarrhea.
- Pneumonia.
- Enlarged lymph nodes in various parts of the body (Lymphadenopathy).
- Appearance of various infectious diseases, whether parasitic, viral, bacterial, or fungal, due to immunodeficiency.

Treatment :-

- The patient should be isolated.
- There is no protective vaccine against the disease yet.
- There are some drugs to treat AIDS but the results are not always curative.

Questions

1. What are the objectives of studying skin diseases in nursing school?
2. How is scabies treated?
3. What are the symptoms of scabies?
4. List the symptoms of head lice?
5. How is head lice transmitted?
6. What is the treatment for genital warts?
7. What are the symptoms of chickenpox?
8. What are the symptoms of herpes simplex?
9. How are bacterial skin infections treated?
10. What are the types of Tinea Capitis? And how are they treated?
11. List the fungal skin diseases that affect the skin surface?
12. What is the treatment for athlete's foot?
13. What are the symptoms of Tinea Versicolor? And how is it treated?
14. List some non-infectious skin diseases and explain one of them?
15. List the symptoms of psoriasis?
16. How is psoriasis treated?
17. What are the symptoms of vitiligo?
18. What is the treatment for acne?
19. Describe the treatment for bullous skin diseases?
20. How is urticaria treated?
21. What are the symptoms of papular urticaria?
22. How is papular urticaria treated?
23. List the types of eczema that affect the skin at different age stages?
24. What is the treatment for eczema?
25. What are the causes of hirsutism in women?
26. How is hirsutism in women treated?
27. What are the types of alopecia that affect hair?
28. What are the treatment methods for alopecia?
29. List the stages of acquired syphilis?

30. What are the stages of congenital syphilis?
31. What are the symptoms of acute gonorrhea in men?
32. What are the symptoms of acute gonorrhea in women?
33. How is gonorrhea treated?
34. What are the symptoms of Acquired Immunodeficiency Syndrome (AIDS)?
35. What is the treatment for AIDS?

Mark (✓) for the correct statement and (×) for the incorrect statement.

1. Scabies is a viral disease transmitted by contact.
2. Scabies is treated by giving Griseofulvin tablets.
3. The incubation period for scabies is five days.
4. Scabies is transmitted by contact.
5. Head lice causes swelling of the lymph nodes in the neck.
6. Chickenpox is transmitted via droplets.
7. Herpes zoster is accompanied by severe pain at the site of the vesicles.
8. Warts are treated by giving antibiotics.
9. Laser can be used to treat warts.
10. Podophyllin solution is used to treat warts affecting the genitals.
11. Chickenpox is accompanied by a slight fever.
12. Tinea Capitis is treated by giving Griseofulvin tablets.
13. Favus (Tinea Favosa) causes damage to hair follicles.
14. Treatment for Tinea Unguium (Onychomycosis) requires a long period.
15. Tinea Versicolor is treated by giving Griseofulvin tablets.
16. Psoriasis is a contagious disease transmitted by contact.
17. Vitiligo is a contagious skin disease.
18. Vitiligo appears as white patches on the body.
19. Vitiligo is spread via droplets.
20. Vitiligo is treated by giving Psoralen tablets (e.g., Meladinine).
21. Psoriasis does not affect the scalp or nails.
22. Blood pressure is monitored periodically in cases of bullous skin diseases.

23. Bullous skin diseases are treated by giving high doses of Cortisone.
24. Treatment for acute urticaria involves giving a subcutaneous Adrenaline injection.
25. Urticaria is treated by giving antiviral drugs.
26. Insect and mosquito bites can lead to papular urticaria.
27. Papular urticaria does not affect children.
28. Eczema is a contagious disease that affects the skin.
29. Eczema affecting children under two years old appears on face.
30. Acne is a result of increased secretion of androgens.
31. Alopecia Areata is a contagious disease that leads to hair loss.
32. Increased secretion of androgens leads to increased hair in women (Hirsutism).
33. Laser is used to treat excess hair.
34. Syphilis is transmitted through sexual contact.
35. The first stage of acquired syphilis is called the chancre (hard ulcer).
36. The incubation period for gonorrhea is from two to three weeks.
37. Gonorrhea is treated with antibiotics.
38. AIDS is a parasitic disease.
39. AIDS is controlled by a protective vaccine against the disease.
40. AIDS is associated with weight loss.

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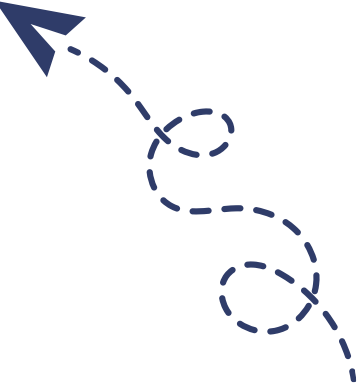
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لمزيد من الملاحظات والكتب الطبية
يُرجى زيارة قناة ثانية تمريضيانو
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